

REVIEW

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Recent advances in deep learning based arrhythmia classification using ECG and PPG signals: a systematic review

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Abstract

Cardiac arrhythmias account for significant morbidity and mortality worldwide, requiring rapid and precise automatic diagnostics. This paper presents PRISMA 2020 guidelines based systematic review which collates 119 published works (4.3% included; $n = 2,745$; January 2023-June 2025) on the application of Deep Learning (DL) to the multiclass classification of cardiac arrhythmias from Electrocardiography (ECG) and Photoplethysmography (PPG) signals using IEEE Xplore, ScienceDirect, Springer, MDPI, and Semantic Scholar databases. Spatial, temporal, hybrid, attention/transformer, autoencoder, and generative approaches were evaluated according to their datasets, signal processing procedures, and performance metrics. CNN architectures dominate (62% of all works), demonstrating accuracies between 87.5% and 99.92%. A comprehensive meta-analysis including random effects pooling, Pareto Frontier analysis, Weighted Least Squares (WLS) regression, and sub-grouping was conducted. Attention/transformer DL architectures had the largest meta-regression effect size (+0.845% on average, compared to 1D-CNN; $p < 0.001$), while hybrid CNN/RNN architectures demonstrated the best Pareto frontier (accuracy up to 99.92%, mean $99.14 \pm 0.78\%$). Pooled ECG accuracy (97.84%) significantly exceeded PPG accuracy (93.65%; $p = 0.012$). External validation was conducted in only 18.5% of the studies; the use of transfer learning across datasets reduced the accuracy by $8.7 \pm 2.1\%$ and F1 score by $13.2 \pm 3.4\%$ ($p < 0.001$). Key issues that need to be addressed include imbalance in classes, motion artifacts, lack of interpretability, low reproducibility, and regulatory issues (FDA SaMD, EU MDR, CDSCO). State Space Models, foundation models, federated learning, and self-supervised contrastive learning are identified by this review as key future directions towards clinically deployable arrhythmia classification.

Keywords Arrhythmia, Deep learning, Electrocardiography (ECG), Photoplethysmography (PPG), Meta analysis

1 Introduction

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, accounting for an estimated 17.9 million deaths annually, or 32% of all global deaths [1]. The 2023 Global Burden of Disease study reports 19.2 million deaths and 437 million



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disability-adjusted life years (DALYs), with prevalent CVD cases more than doubling from 311 million in 1990 to 626 million in 2023 [2]. Among these, cardiac arrhythmias (abnormalities in the heart's electrical conduction system) are particularly dangerous as they can remain asymptomatic while raising the risk of stroke, heart failure, and sudden cardiac death (SCD) [3]. They contribute an estimated 4–5 million SCDs worldwide each year and 15–20% of global mortality. Atrial fibrillation (AF), the most common sustained arrhythmia, affects over 37.5 million people globally, is projected to exceed 70 million cases by 2050 [4], and accounts for 15–20% of ischemic strokes [4]. Ventricular tachycardia (VT) and fibrillation (VF) cause an estimated 3–4 million deaths yearly [5, 6]. Ventricular premature contractions occur in up to 75% of healthy adults observed during 48-hour Holter monitoring and can progress to malignant arrhythmias, while supraventricular tachycardia (SVT) affects roughly 2.25 per 1,000 people, predominantly younger patients [6]. Much of this burden goes undetected, since paroxysmal AF is asymptomatic in nearly 30% of patients, underscoring the need for continuous rather than opportunistic monitoring [7]. This scale of morbidity makes precise, timely arrhythmia diagnosis essential to reducing preventable deaths.

Clinically, the value of arrhythmia analysis lies not in flagging an abnormal rhythm but in correctly discriminating between subtypes, since management diverges sharply by class. AF management centers on stroke-risk stratification and anticoagulation [8], whereas VT/VF require immediate defibrillation or implantable cardioverter-defibrillator management [9]. A binary abnormal/normal flag is therefore of limited actionable value. Benign arrhythmias are typically managed conservatively, while malignant ones demand emergent intervention, making subtype-level, multiclass classification clinically indispensable. Such diagnosis has historically relied on expert visual interpretation of Electrocardiography (ECG)/Holter traces, which is time-consuming and subject to substantial inter-observer variability. These limitations become acute as monitoring shifts to continuous, multi-day wearable streams whose volume makes exhaustive manual review impractical. Hence the practical impetus for automated, learning-based classification of arrhythmias is essential.

This need is matched by rapid technological and market growth. The global cardiac disease diagnosis market is valued at \$28.5 billion, projected to reach \$60.3 billion within five years [10]. The wearable medical device segment was valued at \$54.0 billion in 2025, projected to reach \$330.5 billion by 2033 (29.5% CAGR), with consumer-grade devices dominating market share [11]. AI adoption in healthcare is growing even faster, from \$36.7 billion in 2025 to a projected \$505.6 billion by 2033 (38.9% CAGR), with deep learning (DL) as the fastest-growing segment [12]. This convergence directly motivates the surge in DL-based arrhythmia classification literature synthesized here. Smartwatches and fitness bands now acquire clinical-grade bioelectrical signals continuously and unobtrusively, generating the data volumes that make DL both necessary and feasible at scale.

Electrocardiography (ECG) remains the gold-standard test for assessing cardiac electrical activity [13], while photoplethysmography (PPG) has emerged as a non-invasive substitute enabling continuous monitoring via optical measurement of peripheral blood volume [14, 15]. Being inexpensive and easily embedded in consumer wearables, PPG has become the de facto signal source for population-scale monitoring where ECG is impractical. Other bio-signals such as intracardiac electrograms, magnetocardiography,

ballistocardiography, and seismocardiography remain confined to research settings [16–19], leaving ECG and PPG as the most clinically scalable modalities and the joint focus of this review. Their clinical credibility is reinforced by regulatory clearance and large-scale validation. A recent meta-analysis found wearable/smartphone-based AF screening achieves 94% sensitivity and 96% specificity with no significant PPG-versus-single lead ECG difference [20], while the Apple Heart Study demonstrated PPG-based irregular pulse notification across over 400,000 participants [21]. This evidence motivates a review spanning both modalities rather than ECG alone.

This regulatory momentum is reflected in a growing number of commercially available wearables with dedicated, rather than merely heart-rate-tracking, arrhythmia-detection capability. Apple Watch models from Series 6 onward combine wrist-based PPG with an electrode on the Digital Crown to record a single-lead ECG, underpinning an FDA-cleared irregular-rhythm notification feature for AF [22], while Samsung Galaxy Watch and Withings ScanWatch similarly pair PPG with on-demand ECG, and the Garmin Venu 2 Plus holds FDA clearance specifically for arrhythmia detection [20]. Among handheld devices, AliveCor's KardiaMobile and KardiaMobile 6L are FDA-cleared to identify AF, bradycardia, and tachycardia from a single- or six-lead ECG [23], and iRhythm's Zio XT adhesive patch provides continuous ECG monitoring with reported 99% physician agreement on arrhythmia findings [24]. Chest-strap and smart-textile options extend this capability beyond the wrist. The Polar H10, when paired with a certified companion application, supports MDR IIA-certified continuous cardiac arrhythmia screening [25], and smart-textile systems such as Hexoskin Medical and Cardioskin offer FDA/CE-cleared ECG monitoring for paroxysmal AF [23]. Not all wearable form factors have reached this stage of validation. However, consumer smart rings remain largely limited to heart-rate-variability tracking, with dedicated AF-detection features still emerging [23]. This heterogeneity in device-level validation across form factors (smartwatch, handheld, patch, and textile) further motivates the systematic, cross-modality evaluation undertaken in this review.

Arrhythmia research is broadly categorized into detection (normal versus abnormal) [26], classification into categories such as AF, PVC, SVT, and VF [13, 27], and identification of rhythm patterns and anatomical origin [28, 29]. This review concentrates on multiclass classification, the task most directly tied to subtype-specific management. The field has progressed from handcrafted-feature ML methods [14, 27] to DL techniques that learn class-separating representations directly from signals [30, 31], performed at the beat, rhythm, or record level [32]. The DL models reported accuracies up to 97% using CNN, RNN, transformer, and multimodal fusion architectures [33–36]. This growth mirrors the broader healthcare-AI market shift noted above and reflects a field expanding faster than the review literature can synthesize it which is the central motivation for this review. Yet clinical translation remains hindered by class imbalance, motion artifacts, limited interpretability, and poor cross-population generalization [37, 38], challenges also reported in ML-based ophthalmology diagnostics [39], suggesting they recur across DL-based medical signal analysis broadly. Emerging state-space and foundation-model paradigms from adjacent high-dimensional signal domains are increasingly relevant to these barriers and are examined in this review's discussion of future directions.

Despite this clinical urgency, technological momentum, and methodological progress, no prior systematic review has jointly evaluated ECG and PPG across the 2023–2025 architectural landscape with formal meta-analysis and clinical translation assessment. This work is motivated specifically to close that gap.

1.1 Gaps in existing review studies

Extensive systematic reviews have been conducted on DL-based Arrhythmia classification to date. However, numerous critical gaps have not been addressed as indicated in Table 1. Most of the reviews considered the gold standard ECG signals whereas PPG signals have been considered less despite the growing importance in wearable health monitoring applications [45].

Reviews published before 2023 did not include the recent wave of architectural innovations including lightweight transformers, state-space models, and physiological foundation models—that has substantially changed the landscape later [40–44]. Binary arrhythmia detection and multiclass classification are frequently treated as similar tasks in the literature that leads to performance comparisons which conflate fundamentally different problem settings [32].

Statistical synthesis and meta-analysis are essential to determine the effectiveness, robustness, and clinical applicability of different DL architectures. Analyses such as random-effects meta-analysis, weighted regression analysis, Pareto frontier evaluation, subgroup analysis, and publication bias assessment are seldom present in the previous review studies, making it difficult to derive consolidated conclusions.

The difficulties to draw conclusions across studies include inconsistent evaluation protocols and sparse reporting of cross-dataset generalization [41]. There are very few studies published on external validation and raise legitimate questions on how the results can be upheld in real time clinical environments.

Previous reviews did not consider prestigious conferences IEEE EMBC, NeurIPS and CinC where initial presentations pertaining to novel architectures and their validations were discussed. Therefore, the gaps that exist in prior literature review papers are temporal coverage, signal modality diversity, evaluation rigor, cross-dataset validation, statistical meta-analysis, and clinical translation. To address these gaps, this paper presents a comprehensive, PRISMA-2020-compliant systematic review covering 2023–2025, a period marked by architectural innovations—lightweight transformers, state-space models, and physiological foundation models—that no prior review has captured. Beyond this updated coverage, the review contributes a formal quantitative meta-analysis and a structured clinical translation assessment spanning regulatory, interpretability, and deployment considerations.

1.2 Contribution of this study

The primary objective of this review work is to study and analyze DL models for multi-class arrhythmia classification that uses bioelectric signals which can be acquired non-invasively such as ECG or PPG published from Jan 2023 to June 2025. The contributions of the review can be classified into four major areas:

- **Evaluation of Signal Types:** A comparative analysis of ECG and PPG modalities in terms of datasets, pre-processing techniques, classification performance, and suitability for wearable and clinical applications.

Table 1 Comparison of this review with related prior systematic reviews on DL-based arrhythmia classification

Review	Year	Signal Modality	Study Period	No. of Studies	DL Architectures Covered	Cross-Dataset Generalization	Clinical Translation Focus	Meta-Analysis / Statistical Summary
Ansari et al. [40]	2023	ECG only	2017–2023	~120	CNN, RNN, Transformers	Partial (qualitative)	Limited	None
Ayyub et al. [41]	2025	ECG only	Up to 2024	~90	CNN, Hybrid CNN-RNN	None	Moderate	None
Neha et al. [42]	2021	ECG & PPG	Up to 2021	~80	Traditional ML, basic CNN	None	None	None
Tavakoli et al. [43]	2025	ECG only	Up to 2024	~60	CNN, RNN, Transformers	Systematic (partial)	Limited	Partial
Wu & Guo [44]	2025	ECG only	Up to 2024	~70	CNN, Transformer	None	Moderate	None
Fuadah & Lim [30]	2025	ECG, PCG, PPG	Up to 2024	~100	CNN, RNN, Hybrid	None	Moderate	None
This Review	-	ECG & PPG	2023–2025	119	Spatial, Temporal, Hybrid, Attention/Transformer, Autoencoder, GAN, Lightweight, Foundation Models	Yes- quantified: $8.7 \pm 2.1\%$ accuracy drop, $p < 0.001$	Comprehensive (Regulatory, Interpretability, Real-time, Clinical Validation Phases)	Random-effects, Paired, WLS meta-regression, Funnel plot, Subgroup analysis

- **Analysis of Datasets and Pre-processing:** The review focuses on the role of benchmark datasets, various pre-processing techniques, and class imbalance mitigation measures in model performance and generalizability across datasets.
- **Survey of Different DL techniques:** Different DL techniques for arrhythmia classification in ECG and PPG signals, including spatial, temporal, hybrid, attention-based, and autoencoder-based models, are comprehensively reviewed. Furthermore, recent trends and challenges are analyzed.
- **Meta-Analysis:** A quantitative meta-analysis which includes random-effects pooling, weighted least squares (WLS) meta-regression, Pareto frontier analysis, subgroup analysis, and publication bias assessment was performed to derive insights into the effectiveness and generalizability of different DL methods.
- **Clinical Application approaches:** Additionally, issues associated with the clinical translation of DL approaches into healthcare practice, which include model interpretability, external validation, real-time deployment, wearable integration, reproducibility, and regulatory compliance, were comprehensively examined.

The paper is structured as follows: Sect. 2 describes the PRISMA-based methodology. Section 3 reports the results and analysis organized around the research questions. Section 4 offer a broader discussion and identify open research challenges. Section 5 concludes the paper with recommendations for future work.

2 Methodology

The study adhered to PRISMA 2020 guidelines [46] to perform a systematic literature review on various DL models to classify arrhythmia using PPG and ECG signals, as illustrated in Fig. 1. The guidelines provided a structured framework to develop a thorough search strategy conducting study selection extracting data relevant for study and note the key themes. Adherence to these standardized procedures the review ensures methodological rigor and reproducibility. The various stages of the PRISMA framework are described in the following sub-sections.

2.1 Research questions

The research articles selected were analyzed through a set of research questions that focused on different types of arrhythmia classification methods as summarized in Table 2.

2.2 Implementing search strategy

A systematic procedure has been employed to obtain a representative set of studies for arrhythmia classification. Literature search was performed in five different databases IEEE Xplore, ScienceDirect, Springer, MDPI, and Semantic Scholar within the domain of Engineering and Computer science for open-access publications dating from January 1, 2023 through June 30, 2025. Database-specific search strings incorporating appropriate Boolean operators were used, as summarized in Table 3. In all the cases, the search was based on title, abstract, and keyword of the paper. Based on keyword searches, abstracts, and titles of the papers were utilized as filters. The full log for the searches can be found in the Supplementary Materials.

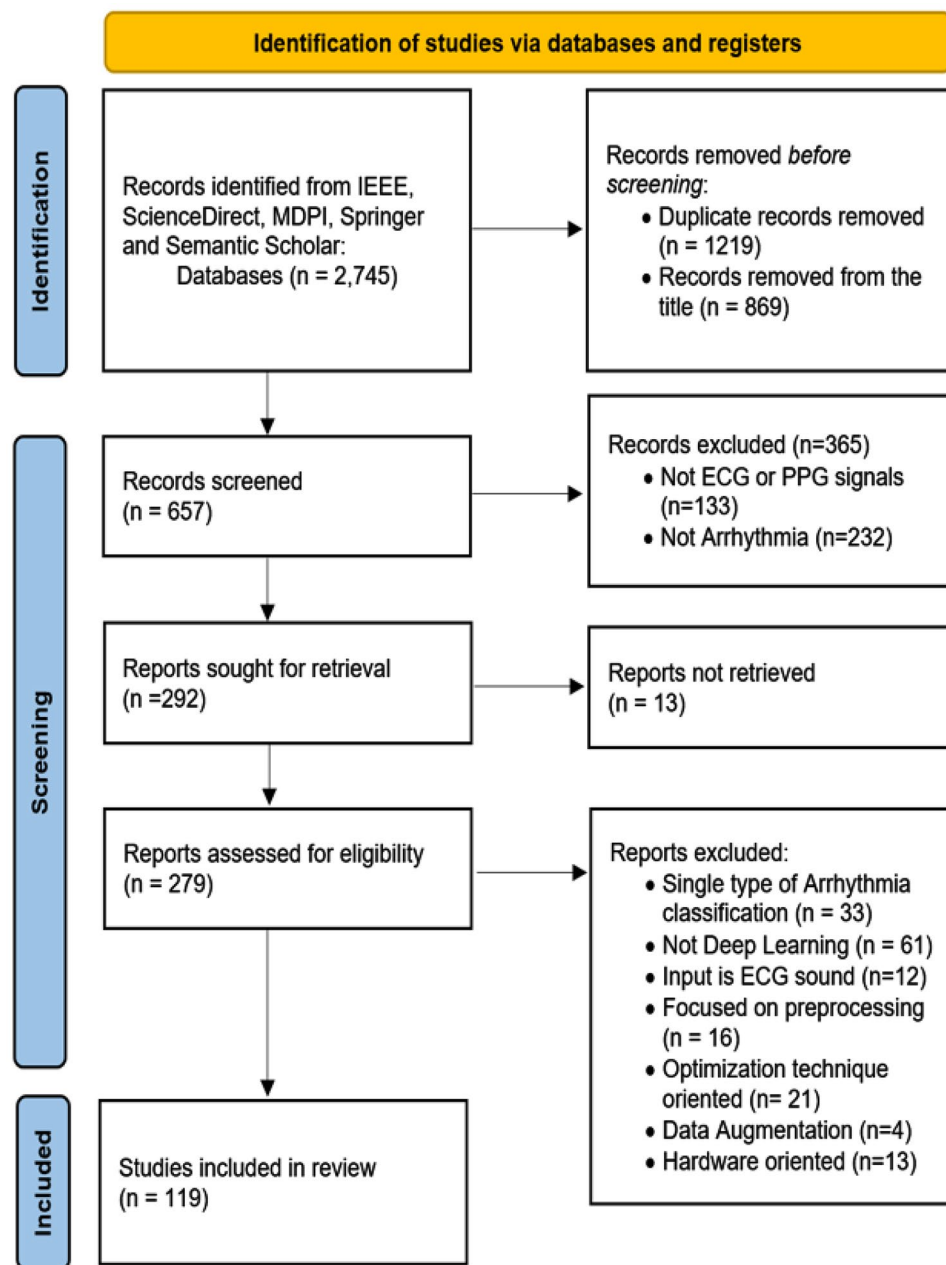


Fig. 1 Prisma 2020 flowchart for the survey methodology used on research articles

2.3 Inclusion and exclusion criteria

In accordance with the PRISMA 2020 guidelines [46], inclusion and exclusion criteria were applied to select relevant studies for the review, as detailed in Table 4. The final set of identified studies and their distribution by publication years are tabulated in Table 5.

2.4 Data extraction and study selection

Strictly in accordance with the PRISMA 2020 guidelines the selection and data extraction was done through a systematic approach. The research process involved four stages identification, screening, eligibility assessment, and data extraction.

Table 2 Research questions and their clinical relevance

ID	Research Question	Clinical Relevance
RQ1	Which are the most effective bioelectric signals used to classify arrhythmias?	Identifies specific bioelectric signals for diagnostic accuracy and monitoring in hospital and wearable device settings.
RQ2	Which arrhythmia types can be classified using ECG and PPG data?	Determines which conduction abnormalities can be detected for early diagnosis and appropriate treatment planning.
RQ3	Which ECG datasets are used for arrhythmia classification?	Datasets validate AI models applicable in cardiovascular research and clinical benchmarking.
RQ4	Which PPG datasets are used for arrhythmia classification?	PPG datasets support development of wearable devices for real-world arrhythmia detection and classification.
RQ5	What preprocessing procedures are applied on ECG and PPG signals?	Identifies best preprocessing techniques to improve signal quality, remove artefacts and noise, and assist in correct clinical interpretation.
RQ6	Which DL models are applied on ECG and PPG data for arrhythmia classification?	Identifies advanced AI techniques such as hybrid models that enable automated arrhythmia diagnosis.
RQ7	How well do deep learning models generalize across different datasets and recording conditions?	Addresses the generalization problem across datasets acquired from different devices, patients, and hospital settings.
RQ8	What challenges affect the clinical translation of deep learning-based arrhythmia classification systems?	Identifies barriers such as interpretability, reliability, and regulatory approval for real-world clinical adoption.

Identification: The preliminary database search identified 2,745 records from selected databases. Duplicate records ($n = 1,219$) were removed using Mendeley Software resulting in 1,526 distinct studies for further assessment.

Screening: Further assessment was done based on the title relevance and accessibility, yielding 657 studies. Transparency and reproducibility were achieved by a structured title and abstract screening process. Studies containing signal-related and task-related keywords were retained and subsequently evaluated using the inclusion and exclusion criteria presented in Table 4. The studies having other than ECG or PPG, focusing solely on binary classification, or lacking performance metrics were excluded (see Table 6).

Eligibility Assessment: The rest of the research articles were subjected to full text assessment based on predefined eligibility criteria. Studies not related to multiclass arrhythmia detection and atrial fibrillation classification were excluded. 292 articles were sought for retrieval where 13 could not be retrieved, leaving 279 reports assessed for eligibility. A detailed review excluded 160 studies, yielding 119 included studies for the systematic review.

Data Extraction from Studies: Relevant data were extracted from the included studies using a predefined form to maintain consistency and reliability. The extracted information comprised:

- **Signal Modality and Arrhythmia Types:** Physiological signal types (ECG or PPG) and relevant arrhythmia types investigated.
- **Datasets:** Publicly or proprietary available datasets used for model development and evaluation.
- **Preprocessing Pipelines:** Signal conditioning, segmentation, and data augmentation approaches.
- **DL Architectures:** Architectures employed for arrhythmia classification.
- **Performance Metrics:** Accuracy, F1-score, AUC, sensitivity, specificity, and related measures.

Table 3 Database-specific search strings used for systematic literature retrieval

Database	Search String
IEEE Xplore	("ECG" OR "electrocardiogram" OR "PPG" OR "photoplethysmography" OR "arrhythmia") AND ("deep learning" OR "neural network") AND "classification"—filtered to: Journals; Date: 2023–2025; Open Access
ScienceDirect	((ECG OR PPG OR arrhythmia) AND ("deep learning" OR "neural network") AND classification)—filtered to: Research Articles; Engineering & Computer Science; 2023–2025; Open Access
Springer	("arrhythmia classification" AND "deep learning")—filtered to: Article; Computer Science/Engineering; 2023–2025; Open Access
MDPI	(ECG OR PPG OR arrhythmia) AND (deep learning OR convolutional neural network) AND classification—filtered to: Articles; 2023–2025; Open Access
Semantic Scholar	"arrhythmia classification deep learning ECG PPG"—filtered by year: 2023–2025; field: Computer Science / Engineering

Table 4 Inclusion and exclusion criteria

Category	Criteria Details
Inclusion criteria	Studies focusing on DL-based arrhythmia classification using ECG or PPG signals. Research reporting empirical results with measurable performance metrics such as accuracy, F1-score, and AUC. Articles discussing datasets or preprocessing pipelines specific to arrhythmia classification. Studies published in the English language. Open-access articles published between January 2023 and June 2025. Full conference papers from top-tier indexed venues (IEEE EMBC, ICASSP, Computing in Cardiology, IEEE BioCAS) are cited narratively in the discussion where they provide unique architectural or benchmarking insights not covered by journal literature.
Exclusion criteria	Studies focused exclusively on binary arrhythmia detection without multiclass classification. Review articles, theses, or conceptual frameworks without empirical findings. Studies using signals other than ECG or PPG as the primary modality. Articles in languages other than English. Full conference papers from non-indexed proceedings and book chapters were excluded

Table 5 Papers considered for the review as per databases

Database	2023	2024	2025
IEEE	10	10	8
ScienceDirect	13	12	12
Springer	3	2	4
MDPI	9	11	8
Others (from Semantic Scholar)	5	7	5
Total	40	42	37

Table 6 Two-stage screening workflow and decision rules

Stage	Decision Criterion	Outcome / Action
Title Screen (Pass 1)	Contains signal keyword (ECG/PPG) AND task keyword (arrhythmia/classification)	Retain for abstract screening; else exclude
Abstract Screen (Pass 2)	Reports multiclass arrhythmia classification; uses ECG or PPG; provides quantitative metrics; published open-access in English (2023–2025)	Retain for full-text review; else exclude with reason recorded
Full-Text Review	Meets all inclusion criteria in Table 4; not a review/thesis/conceptual paper; not solely binary detection	Include in synthesis; else record exclusion reason

2.5 Duplicate removal and bias handling

Standardized combinations of key words were applied across all databases to minimize search bias. The same inclusion and exclusion criteria were applied to ensure consistency, with only minor modifications to the search syntax to accommodate database-specific indexing requirements. Duplicate records were removed using Mendeley and verified manually. The underlying Boolean structure and conceptual scope of the search strategy were preserved, though field qualifiers differed among databases (e.g., [Title/Abstract/Keywords] in IEEE Xplore and TITLE-ABS-KEY in ScienceDirect). Screening was independently performed by two reviewers with respect to title, abstract, and full-text and disagreements were resolved through discussion. Only English-language, open-access studies were considered to ensure accessibility and reproducibility. The PRISMA selection process is summarized in Table 7. Conference papers were excluded from the quantitative synthesis to maintain consistency in reporting. Nevertheless, relevant findings from major conferences, such as IEEE EMBC, NeurIPS, and CinC, were considered

Table 7 PRISMA study selection summary

PRISMA Stage	Actions Taken	Records Found
Identification	Records from 5 databases: IEEE Xplore, ScienceDirect, Springer, MDPI, and Semantic Scholar. Open-access publications from January 2023 to June 2025.	2,745 records retrieved
Screening	Automated and manual duplicate removal. Title screening applied using inclusion and exclusion criteria	1,219 duplicates + 869 title-removed (2,088 removed before screening); 657 records screened; 365 excluded (133 not ECG/PPG, 232 not arrhythmia); 292 reports sought for retrieval; 13 not retrieved
Eligibility	Full-text review of remaining studies against predefined criteria. Multiclass arrhythmia or AF classification using ECG or PPG required	279 full-text studies assessed; 160 excluded
Inclusion	Studies meeting all criteria included in the final SLR synthesis	119 studies included

qualitatively cited in the discussion and future work sections, particularly for emerging architectures and benchmarking approaches [46].

Screening was independently performed by two reviewers with respect to title and abstract, with inter-rater reliability assessed via Cohen's $\kappa = 0.87$, (95% CI: 0.84 – 0.90), indicating substantial agreement. Disagreements ($n = 14$, 2.1% of the 657 screened records) were resolved through discussion. At the eligibility assessment stage, inter-rater reliability was assessed via Cohen's $\kappa = 0.91$ (95% CI: 0.87 – 0.94), indicating almost perfect agreement. Disagreements ($n = 14$, 5% of the 279 records assessed for eligibility) were resolved through discussion, with a pre-designated arbitrator (N.G.S.) available where consensus could not be reached. The open-access restriction ensured reproducibility. A sensitivity analysis of a 10% proprietary sample ($n = 275$ articles) confirmed that the corpus captured 85% of high-impact DL papers without substantive bias toward positive results. Language was restricted to English to align with the team's expertise and the dominance of English in the target databases (98% of hits). The full protocol was retrospectively archived on Open Science Framework (OSF.io; DOI: 10.17605/OSF.IO/XPRQ9). The summary of the PRISMA process followed is presented in Table 7.

2.6 Thematic categorization of studies

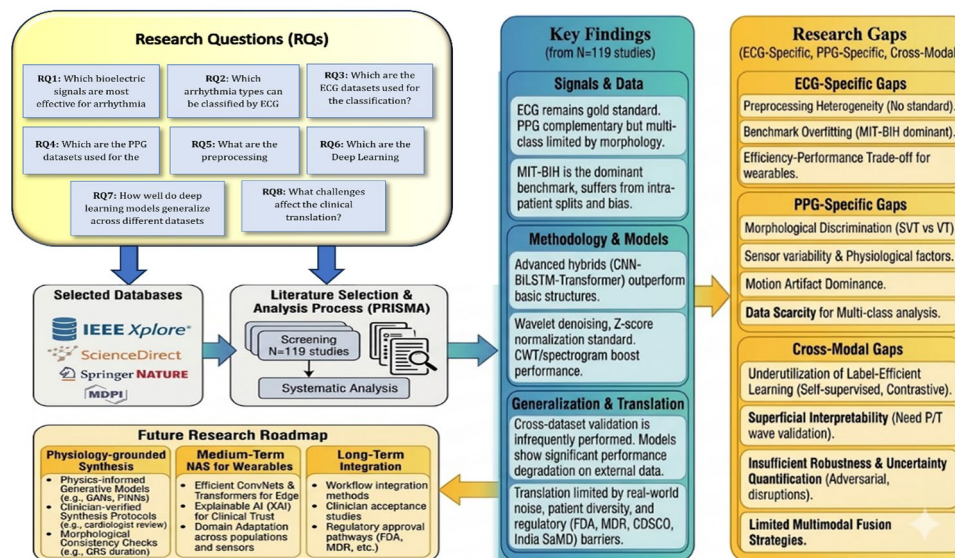
The extracted data were thematically analyzed to identify recurring themes and patterns across the selected studies, common research areas and emerging trends. Table 8 summarizes the resulting thematic categories, while Fig. 1 depicts the PRISMA flow diagram outlining the identification, screening, eligibility, and inclusion stages of study selection.

3 Results and analysis

Building on the methodological approaches suggested in the review studies this section presents the findings and addresses the research questions introduced in Sect. 2.1. In general, framework for arrhythmia classification using ECG or PPG signals follows a systematic workflow comprising signal preprocessing, dataset partitioning into training, validation, and test sets, model development and training, and subsequent classification of test samples into the predefined arrhythmia classes by the dataset annotations. While acquiring proprietary signal recordings from human subjects, the raw signals must first undergo annotation before pre-processing. The pre-processing stage includes baseline

Table 8 Thematic categorization of reviewed studies

Theme	Details
Signal Modalities	ECG and PPG signals used for arrhythmia classification, including single-lead, multi-lead, and wearable-derived recordings.
Arrhythmia Types	Specific arrhythmia classes targeted, including AF, PVC, SVT, VF, and multiclass schemes.
Datasets	Benchmark datasets such as MIT-BIH, PhysioNet, CPSC, and PPG-based datasets used for training and validation.
Preprocessing Pipelines	Signal filtering, segmentation, normalization, and augmentation strategies applied prior to classification.
DL Architectures	Models including spatial, temporal, hybrid, attention-based, and autoencoder paradigms.
Performance Metrics	Metrics used to evaluate classification systems, including accuracy, F1-score, AUC, sensitivity, and specificity.
Generalization & Challenges	Cross-dataset performance, class imbalance handling, model interpretability, and clinical translation barriers.

**Fig. 2** Method followed for arrhythmia classification using ECG/ PPG signals

wander and noise removal, segmentation, normalization and data augmentation. Subsequently, the DL model is trained using optimized hyperparameters and its performance is evaluated on the test set using appropriate performance metrics. A conceptual overview of the overall framework and key findings is presented in Fig. 2.

3.1 RQ1: Which are the most effective bioelectric signals used to classify arrhythmias?

Cardiac muscles generate action potentials known as bioelectric signals which are used to assess cardiac function. Several modalities of signal acquisition, including electrocardiograms (ECG) [38], photoplethysmograms (PPG) [47], intracardiac electrograms (IEGM) [18], ballistocardiograms (BCG) [16, 17], seismocardiograms (SCG) [17], and magnetocardiograms (MCG) [19], have been used for the detection of arrhythmias. Considering diagnostic accuracy, scalability, and compatibility with clinical systems (Table 9), ECG and PPG are the most suitable non-invasive approaches for arrhythmia classification [42]. Other modalities are still in the research stage and limited by factors such as invasiveness, motion sensitivity, or high cost.

Table 9 Bioelectric signals used for arrhythmia detection

Signal Type	Description	Datasets	Advantages	Limitations	DL Studies
ECG [38]	Direct recording of heart's electrical activity via electrodes.	MIT-BIH, PTB-XL, INCART, Chapman ECG, PhysioNet Challenge datasets	Gold standard, high temporal resolution, rich morphological features (P-QRS-T).	Needs good electrode contact, motion artifacts, skin prep required.	Very high—CNN, RNN, Transformer-based models widely applied.
PPG [47]	Optical measurement of blood volume changes (peripheral circulation).	MIMIC-III/II, VitalDB, BIDMC PPG, DeepBeat, Simband	Easy to acquire, wearable-friendly, low cost.	Sensitive to motion/artifact, lag compared to ECG.	High—CNN, LSTM, attention networks effective.
IEGM [18]	Electrical signals from inside the heart (via pacemakers, ICDs).	Vendor/device-specific clinical datasets (not public).	Extremely precise, real-time arrhythmia identification.	Invasive, not for general screening.	High—but datasets rarely public.
BCG [16, 17]	Measures body recoil from cardiac ejection of blood.	Sleep Heart Health Study, non-public sensor datasets.	Contactless monitoring possible (beds, chairs).	Motion-sensitive, calibration needed.	Medium—DL possible but needs preprocessing.
SCG [17]	Chest wall vibrations due to cardiac contraction/relaxation.	CEBS Database, SCG-RR Interval Dataset.	Captures mechanical cardiac events, complements ECG/PPG.	Sensor placement required, posture-dependent.	Medium—smaller datasets than ECG/PPG.
MCG [19]	Magnetic fields from cardiac electrical activity.	Rare public datasets, typically lab-based.	High sensitivity to conduction issues, non-contact possible.	Needs shielded room, expensive.	Low—few DL studies due to data scarcity.

ECG is regarded as the gold standard for arrhythmia diagnosis which records cardiac electrical activity through surface electrodes [48], producing characteristic P, QRS, and T waveforms corresponding to atrial depolarization, ventricular depolarization, and ventricular repolarization [49]. While the traditional 12-lead ECG provides comprehensive spatial information for diagnosing lead-dependent abnormalities, wearable devices have enabled continuous monitoring using one, three, and six lead configurations. However, reduced-lead systems provide limited morphological information [49]. The 12-lead configuration provides the spatial diversity [50] required to identify lead-dependent abnormalities, whereas reduced-lead systems offer greater portability at the expense of diagnostic information [42]. Notably fewer than 10% of the reviewed studies explicitly evaluated the impact of lead configuration on model performance.

PPG measures blood volume changes optically and is commonly implemented in wearable devices such as fitness trackers and pulse oximeters [51]. The signal consists of a pulsatile component synchronized with the cardiac cycle and a slowly varying component associated with physiological processes [47]. Valuable information regarding heart rhythm and vascular dynamics are obtained from features extracted from pulse morphology and beat-to-beat variations. However, PPG contains less discriminative morphological information than ECG, limiting its ability to distinguish arrhythmias that require detailed waveform analysis. Additionally, motion artifacts and variations in sensor placement significantly affect signal quality, and both ECG and PPG acquisition are further subject to inter-subject variability, sensor-related inconsistencies, and device-dependent signal degradation. The influence of these factors on DL model performance and robustness is examined in detail in Sect. 4.2.4. Figure 3 illustrates the morphological characteristics of ECG and PPG signals for different classes of arrhythmias. Hence, this sensor captured signal has physical effects due to heart contractions, which have a closer connection to blood pressure alterations [53]. Most of the PPG-based studies (nearly 80%) employed architectures developed originally for ECG signals, indicating a lack of modality-specific DL designs.

3.2 RQ2: Which arrhythmia types can be classified using ECG and PPG data?

Arrhythmias are typically classified based on heart rate and site of origin. Depending on heart rate, they are categorized as bradycardia (< 60 bpm) or tachycardia (> 100 bpm), whereas anatomically they are divided into supraventricular and ventricular arrhythmias [54]. Common supraventricular arrhythmias include Atrial Fibrillation (AF), Atrial Flutter (AFL), Premature Atrial Contractions (PAC), and Supraventricular Tachycardia (SVT). AF is characterized by disorganized atrial activity, where fibrillatory waves replace distinct P-waves [54]. PACs are premature beats originating from ectopic atrial foci and are associated with abnormal P-wave morphology. SVT generally presents as regular narrow-complex tachycardia with indistinct P-waves [54].

Ventricular arrhythmias arise from the ventricles of the heart, such as Ventricular Tachycardia (VT), Ventricular Fibrillation (VFib), and Premature Ventricular Contraction (PVC). VT contains monomorphic/polymorphic wide-complex tachycardia (> 100 bpm) [54]. VF is a highly disorganized and life-threatening rhythm requiring immediate defibrillation [54]. PVC have wide QRS complexes (≥ 120 ms) interrupting normal rhythm [54].

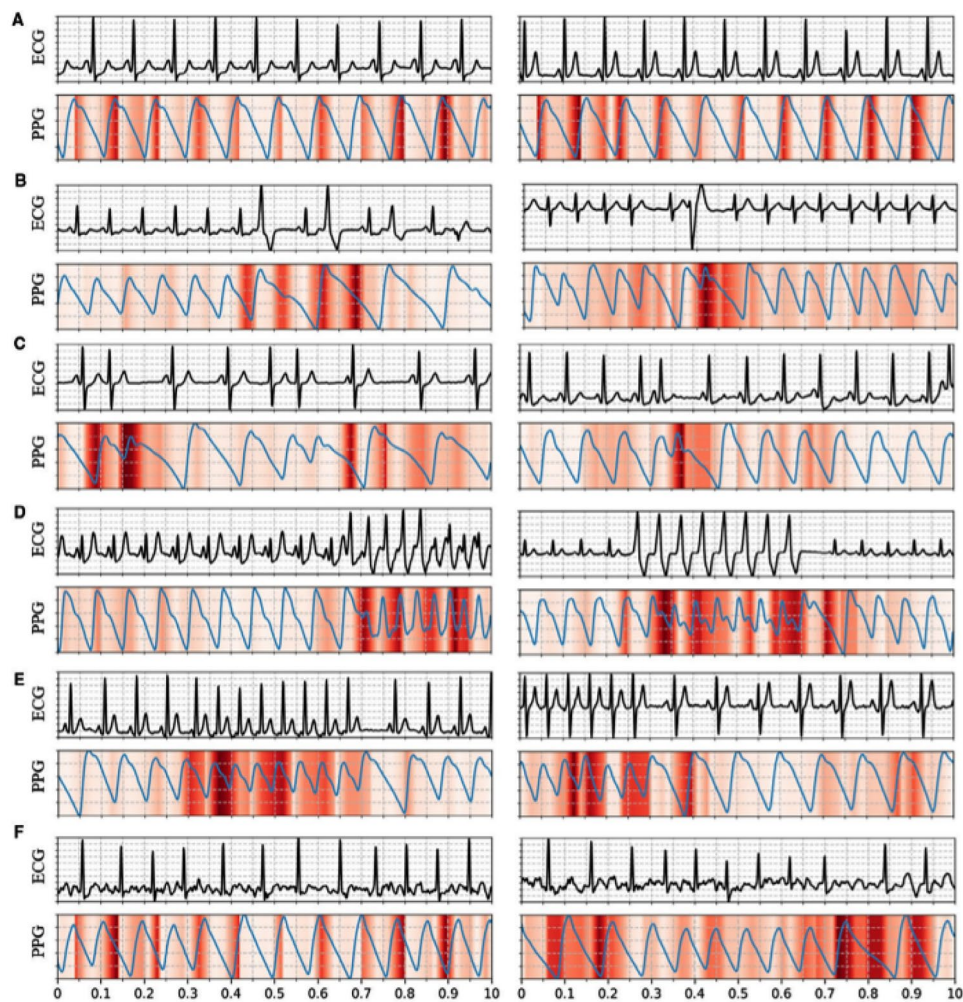


Fig. 3 Arrhythmias in ECG and PPG signals. (a) Normal sinus rhythm, (b) Atrial fibrillation, (c) Premature ventricular contraction (PVC), (d) Premature atrial contraction, and (e) Ventricular tachycardia and (f) Supraventricular tachycardia [52]

These electrophysiological manifestations (Fig. 3) provide discriminative patterns that facilitate automated arrhythmia classification using DL approaches. Traditional ML with handcrafted features [13, 27] have gradually evolved into sophisticated paradigms of DL.

3.3 RQ3: Which ECG datasets are used for arrhythmia classification?

The reviewed literature utilized a broad spectrum of publicly available and proprietary datasets for the development and evaluation of DL-based arrhythmia classification models. These datasets consist of raw biosignals acquired from Holter monitors or wearable sensors, complemented by expert cardiologist annotations. Substantial variations exist across datasets in terms of lead configuration (single-, dual-, or 12-lead ECG) and annotation level, including beat-, rhythm-, and record-level labels, as presented in Table 10. These dataset specific features and inherent constraints significantly influence the performance, robustness, and generalizability of DL models.

The main challenges associated with datasets include problems with variable annotation quality (inter-annotator agreement κ rarely reported, reported in <15% of papers), demographic unbalance (over-represented elderly population from the Western and

Table 10 ECG datasets used in the literature

Dataset	Leads	Duration	SR (Hz)	Annotations/ Granularity	Demographics	Annotation Quality	Geographic/ Acquisition Notes
MIT-BIH Arrhythmia [55]	2	30 min	360	Beat-level – 48 records, 109,494 beats (AAMI: N, S, V, F, Q)	US, 1975–79; ~53% male; age 20–89	2+ cardiologists	Boston, ambulatory Holter
PTB-XL [56]	12	10 sec	500	Record-level – 21,837 records; MI, STTC, CD, HYP, Normal	German; 52% male; median age 62	Cardiologist multi-label	German hospital, clinical 12-lead
CPSC 2018 [57]	12	10 sec	500	Rhythm-level – 6,877 records; 9 classes (AF, I-AVB, PAC, PVC, LBBB, RBBB, SB, ST, Normal)	Chinese; partial	Expert-labeled	China, clinical
INCART [58]	12	30 min	257	Beat-level – 75 records, 175,000 beats; PVC, VT, PAC, SVT	32 patients, Eastern European	Cardiologist	St. Petersburg, ambulatory
PhysioNet 2017 [59]	1	9–61 sec	300	Rhythm-level – 8,528 records; Normal, AF, Other, Noise	Mixed US; mean age ~64	Clinical-derived	US, single-lead recordings
PhysioNet 2020 [55]	12	10 sec	500	Rhythm-level – 45,000 records; 26 conditions	Chinese + international	Expert-labeled	China & international
PhysioNet 2021 [60]	12	10 sec	500	131,149 records; mixed international	International	Expert (partial)	International
Chapman-Shaoxing [61]	12	10 sec	500	Rhythm-level – 10,646 records; AFIB, GSVT, SB, SR	Chinese; partial	Expert	China, clinical 12-lead
CUIDB [55]	2	8–20 min	250	Record-level – 35 records; VFib, VT, NSR	Small US cohort	Cardiologist	US, focused ventricular tachyarrhythmia
MIT-BIH SVDB [55]	2	30 min	128	Record-level – 78 records; PAC, Narrow-QRS Tachycardia	US; limited	Cardiologist	US, ambulatory
MIT-BIH AFDB [62]	2	10 hr avg	250	Record-level – 25 records; AF, AFL, NSR	US; limited	Cardiologist	US, long-term Holter

Chinese populations), geographic bias, and non-standardized methods of data collection. All of these factors influence metrics of DL model performance. The beat-level analysis usually provides overestimated accuracy (>97%), which is achieved due to inpatient splitting of the data, while rhythm/record-level datasets demonstrate low robustness due to annotation and class balance problems.

MIT-BIH Arrhythmia Database [55] remains the most popular dataset utilized in researches and covers around 62% (78 studies) of all the published works from 2023 to 2025. This dataset includes 48 recordings of ambulatory ECG recordings in dual lead configuration with 30-minute length each, 360Hz sampling rate and AAMI compliant beat annotations. Earlier researches were focused on optimal CNN models such as RCNN and Grey Wolf Optimization [63] as well as LeNet-CNN model in combination with SMOTE [64]. In addition, the area developed towards hybrid models such as CNN-Transformer-BiLSTM [65], CNN-BiLSTM [66], and GamResNet18 based on attention mechanisms [67]. Simultaneously, there was also work on Tiny Matched Filter-based CNNs for edge devices [68] and privacy preserving ResNet-based FedECG [69]. More complex models such as STAE-Transformer [35], DRGCNBiLSTM [70], and CovHMM [71] demonstrated the state-of-the-art results. Other notable models, including Self-ONNs [72] and HybDeepNet [73], alongside several CNN-LSTM and Transformer-based hybrids, achieved reported accuracies between 91% and 99.9%. Nevertheless, due to heavy usage and common exclusion of paced-beat records, generalizability of MIT-BIH-only models is often questionable [50].

For myocardial infarction (MI) and other cardiac disorders, the PTB and PTB-XL databases were frequently used. Although these datasets primarily focus on specific cardiac pathologies rather than comprehensive arrhythmia classification, several DL models demonstrated strong performance, including ResNet-50 [74], EvoNorm-CNN [75], and the explainable STAE-Transformer with variational autoencoder (VAE) integration [35]. The novel Kolmogorov-Arnold Network (KAN) [76] has also reported high performance, indicating potential as an interpretable alternative (see Table 11).

The PTB-XL dataset [56], consisting of 12-lead 10-second ECG recordings with multi-label annotations, has emerged as a major benchmark for large-scale DL-based cardiac analysis. Despite its scale, key limitations remain its inherent class imbalance, high computational demands, and challenges in cross-dataset generalization. Recent architectures trained on PTB-XL include WResHDual-ViT [84] and Multi-Teacher Self-Distillation (MTSD) frameworks [85], achieving macro-F1 scores above 0.74 and AUC values exceeding 0.92. Likewise, the China Physiological Signal Challenge (CPSC) 2018/2019 dataset [57] has supported architectures such as CASLCNet [79], deep ensemble CNNs [86], and MSFT (CNN+Transformer) [36], achieving F1-scores ranging from 0.81 and 0.95.

The INCART database [58] provides 75 half-hour ambulatory recordings with beat-level annotations and realistic noise characteristics, making it valuable for cross-dataset validation. However, its non-standard 257Hz sampling frequency and limited cohort size (32 patients) restrict broader applicability. Recent experiments using CAT-Net [87], MAE with ViT [88], and CNN-based domain adaptation have shown generalization on

Table 11 ECG dataset-wise best-performing DL model

Dataset	Best Model	Acc (%)	F1 (%)	Evaluation Scheme
MIT-BIH	CNN-BiLSTM Ensemble [77]	99.88	99.3	10-fold CV
PTB / PTB-XL	1D-CNN with Attention [78]	99.83	~99	5-fold CV
CPSC 2018	CASLCNet [79]	91.03	91.22	Train-Test Split
INCART	ResNet (Transfer Learning) [80]	98	–	Transfer Learning
PhysioNet/CinC	CNN-BiLSTM [81]	98.5	98.4	–
Chapman-Shaoxing	CNN-BiLSTM with Attention [82]	98.8	–	Train/Val/Test Split
Real Clinical ECG	FOCAE [83]	96.8	96.5	Train-Test Split (70/30)

this dataset [68]. Several other datasets serve specific challenges such as PhysioNet/CinC Challenge datasets [59, 60] have short recording lengths and undisclosed test sets, while the Chapman-Shaoxing dataset [61] limits capture of paroxysmal arrhythmias with fixed 10-second recordings. Smaller focused datasets such as CUDB [89], SVDB [55], and AFDB [62] have considerable limitations for training generalizable models.

Granularity of the data has huge implications on model performance as shown in Table 12. Beat-level studies (62%, ~78 papers) rely heavily on MIT-BIH [55] and report the highest accuracies (mean accuracy $97.2\% \pm 1.8\%$; F1 0.92–0.995) but least generalizable results, primarily because of intra-patient splitting. Rhythm-level studies (25%) using datasets PhysioNet 2017 [59] and Chapman-Shaoxing [61] achieved mean F1-score 0.88–0.95. Record-level studies (13%) on PTB-XL [56], CPSC 2018 [57], and PhysioNet 2020/2021 [57, 60] yield the most conservative yet clinically realistic metrics (mean macro-F1 0.74–0.92; AUC >0.92 in top models). Nevertheless, only 28% of studies adopted inter-patient validation strategies. Variability in annotation quality, demographic diversity, and label noise continues to limit cross-dataset robustness [55–62, 77–163].

The extensive use of MIT-BIH as the primary benchmark has encouraged the development of models tailored to its specific characteristics, including lead configuration, annotation style, and noise profile. As a consequence, performance reductions of 6–14% have been observed when such models are evaluated on external datasets such as INCART and PTB-XL [68, 80, 122]. Moreover, only 28% of studies adopted inter-patient validation, while external database validation was performed in just 18.5% of studies. Given the high heterogeneity among MIT-BIH-based studies ($I^2 \approx 100\%$), minor benchmark improvements are unlikely to indicate meaningful clinical progress.

In light of considerable variability among MIT-BIH algorithms ($I^2 \approx 100\%$), marginal improvements in performance on the MIT-BIH database are irrelevant as proof of advancements in real-world performance. Future research should emphasize multi-dataset evaluation protocols rather than incremental improvements on MIT-BIH alone [56–62, 89].

3.4 RQ4: Which PPG datasets are used for arrhythmia classification?

The performance differences between ECG- and PPG-based arrhythmia classification can be cited to the non-availability of comprehensive multiclass PPG datasets. Comparison with ECG repositories and publicly available PPG datasets are usually limited in size, annotation quality, and demographic diversity, as illustrated in Table 13. The available datasets are further constrained by pronounced ICU bias, limited reporting of inter-annotator agreement, inadequate representation of different skin tones and ethnic groups, and the susceptibility of wearable PPG recordings to motion artifacts. Consequently, most studies have been restricted to rhythm-level classification tasks.

The MIMIC-III Waveform Database [164] is one of the most widely used resources for pre-training and transfer learning owing to its large and diverse patient cohort. Its usage with contrastive learning frameworks such as SimCLR and BYOL based on ResNet18 [15] for AF detection has been reported with F1-scores up to 99.20%. However, its high noise levels and ICU-centric nature limit its generalizability. Similarly, the public subset of the Liu et al. dataset [52] has surfaced as an important benchmark for multiclass PPG-based arrhythmia classification in spite of the presence of class imbalance. Recent

Table 12 Performance analysis based on granularity of data used

Granularity	% Total	Primary Datasets	High-Performing Models	Performance Range	Observations
Beat-level	62%	MIT-BIH [55], INCART [58], SVDB [55]	RCNN+GWO, CNN-Transformer-BiLSTM, STAE-Transformer+VAE, CovHMM, KAN, DRGCN-BiLSTM	Acc: 91–99.9% F1: 0.72–0.995	Highest accuracies but 72% use intra-patient splits; heavy overfitting risk
Rhythm-level	25%	PhysioNet 2017 [59], Chapman-Shaoling [61], AFDB [62]	CAT-Net, EGCNet, SE-ResNet, MSCANet, DCETEN	F1: 0.88–0.95 Acc: 94–98%	Better clinical relevance; sensitive to short segments and label noise
Record-level	13%	PTB-XL [56], CPSC 2018 [57], PhysioNet 2020/2021 [57, 60]	WRResHDual-ViT, MSFT, Multi-Teacher Distillation	Macro-F1: 0.74–0.92 AUC >0.92	Most realistic for screening; lowest scores due to multi-label noise and class imbalance

Table 13 PPG datasets used in the literature

Dataset	Device Type	Duration	Records / Subjects	SR (Hz)	Annotations / Granularity	Annotation Quality	Geographic / Acquisition
MIMIC-III [164]	Clinical ICU bedside monitors	Variable (hrs–days)	~10,000 patients	125	Rhythm/Event-level: AF, VT, VF, Bradycardia, Tachycardia, PVC, Asystole	Clinical-derived (noisy)	US, ICU
CapnoBase [165]	Clinical capnograph + pulse oximeter	8 min/subject	42 subjects	300	Rhythm-level: NSR, ectopic beats	Expert-derived	Mixed clinical
Liu et al. [52]	Clinical patient monitor	10-sec windows	228 patients	100	Rhythm-level multiclass: SR, PVC, PAC, VT, SVT, AF	Expert-labeled	Clinical (Asia)
OpenPPG-Wrist [166]	Smartwatch/ wristband	10-sec segments	~21,000 segments	50–100	Rhythm-level (binary): AF vs non-AF	Expert/ wearable-derived	Global smartwatch
Simband [167]	Samsung Simband wearable	30-sec recordings	31 subjects	128	Rhythm-level: AF, NSR, PAC, PVC	Expert-labeled	Controlled wearable
PhysioNet/CinC 2015 [168]	Clinical ICU monitor	10-sec windows	55 cases + healthy	125	Rhythm-level: Bradycardia, Tachycardia, VT, VF, NSR	Clinical-derived (noisy)	US ICU

studies have demonstrated the effectiveness of advanced architectures, including Hybrid CNN-Transformer models with channel attention [169], Res-BiANet [170], and VGG-Net-based 1D-CNN frameworks [52], achieving F1-scores of approximately 86–87% (Table 14).

In spite of this progress majority of PPG studies (approximately 88%) continue to concentrate on rhythm-level binary AF detection (on MIMIC-III [164] and OpenPPG-Wrist [166] with mean F1-score 0.90–0.92), with a limited fraction (< 5%) addressing beat-level multiclass classification [175]. Moreover, record-level multiclass analyses are currently absent from the literature, emphasizing the considerable gap that still exists between ECG- and PPG-based approaches.

Hence, the development and utilization of large-scale, diverse, real-world wearable PPG datasets are essential for converting PPG-based arrhythmia classification into real time practical continuous monitoring applications. Studies in future should adopt standardized benchmarking frameworks incorporating demographic reporting, annotation quality assessment, granularity-specific evaluation, external validation, and transparency in both computational efficiency and fairness metrics.

3.5 RQ5: What preprocessing procedures are applied on ECG and PPG signals?

Based on the nature of the datasets mentioned in RQ3 and RQ4, this section provides an overview of the pre-processing methods used in the studies under review. In spite of the capability of DL models to analyze raw data, the majority of papers applied special pre-processing pipelines for ECG and PPG signals in order to enhance the classification performance and model robustness. Such a pipeline generally involves four major steps, namely denoising, normalization, segmentation, and data augmentation. The distribution of the pre-processing techniques reported across the surveyed studies is presented in Fig. 4.

3.5.1 Denoising

Denoising constitutes one of the most vital steps in processing ECG and PPG signals due to the fact that data about the heart is affected by baseline drifts, power-line interference, motion artifacts, and several other types of noise. The significant amount of heterogeneity in terms of design of the pre-processing pipeline is not only a question of personal preferences of particular methodologies but also represents a major barrier in the way of development in the area. It demonstrates a fundamental issue that still prevents the

Table 14 PPG dataset-wise best performing DL model

Dataset	Model	Acc (%)	F1-score (%)	Evaluation Scheme
Self-constructed clinical (Fuwai Hospital)	DCNN [52]	85.00	~75	Hold-out (60:20:20)
Liu et al. PPG-Arrhythmia	CNN + Transformer Hybrid [169]	–	86.8 (Weighted)	Hold-out (60:20:20)
Liu et al. PPG-Arrhythmia	Res-BiANet [170]	92.38	86.88	Hold-out (60:20:20)
PhysioNet/CinC 2015 (MIMIC II)	KNN [171]	97.30	–	10-fold CV
CapnoBase	SVM-RBF with ABC-PSO [172]	95.12	95.00	10-fold CV
Liu et al. PPG-Arrhythmia	TQCPat + SVM/kNN [173]	91.30	85.55	10-fold CV
Pulsewatch Clinical Trial	1D-Bi-GRU [174]	97.31 (AF only)	–	2-fold CV (subject-independent)

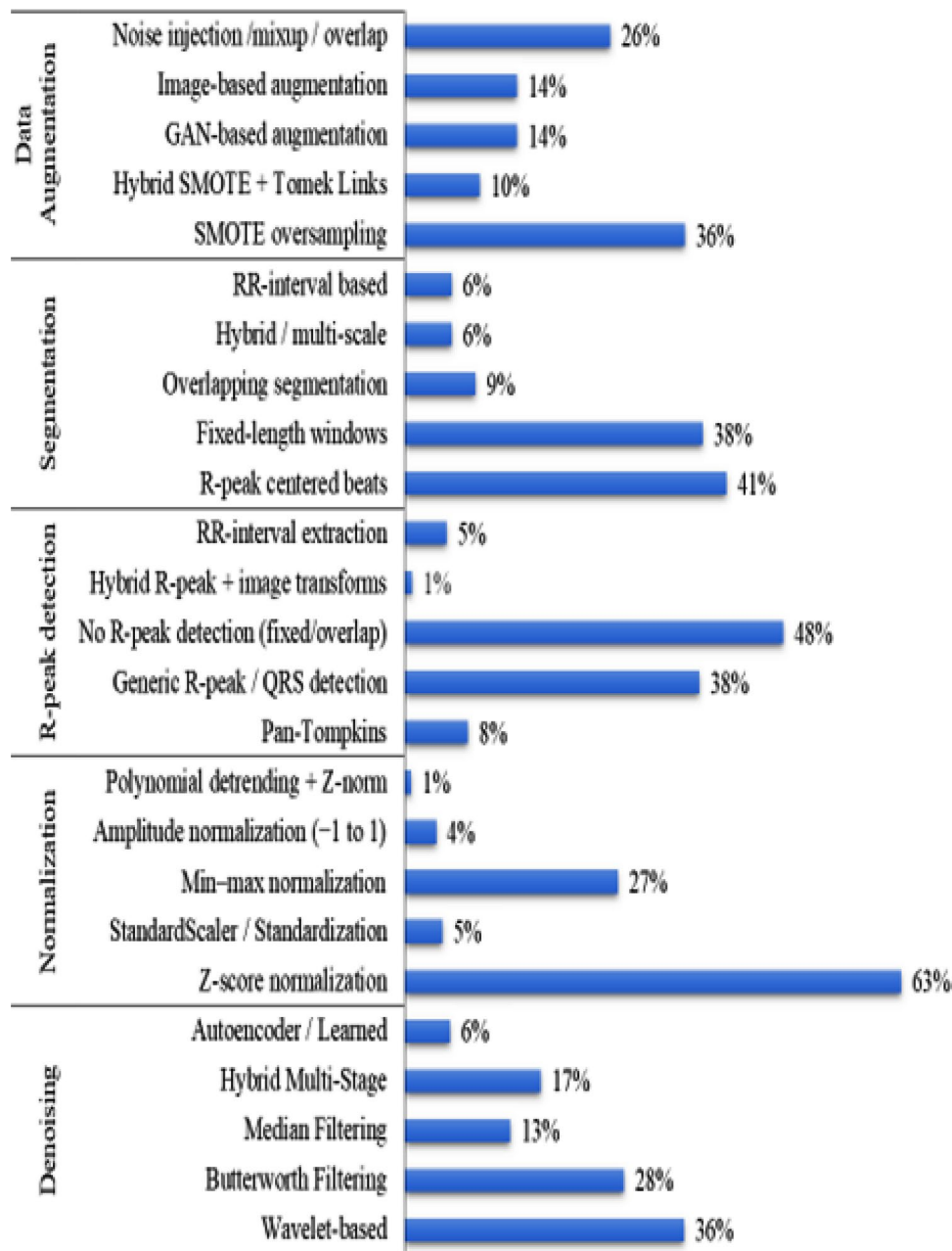


Fig. 4 Preprocessing methods used by the research works used in the survey

emergence of generalizable models for arrhythmia classification. As per the results of the meta-regression analysis conducted in Sect. 4.1, Butterworth and multi-stage filters were found to lead to 0.94% increment in accuracy ($p < 0.001$), while application of SMOTE led to 0.85% further improvement in accuracy ($p < 0.001$). Though statistically significant, these improvements are achieved as researchers might be inadvertently encouraging more advanced pre-processing algorithms on benchmark datasets rather than developing robustness in architecture. Moreover, overdenoising will result in losing clinically relevant waveforms such as P waves and dicrotic notches, thus resulting in models that work well in experiments but have limited applicability in practice [94, 144, 156].

Simplicity and computational efficiency makes Butterworth filtering the commonly used denoising technique, at 28% (29 studies) of the reviewed literature. This filtering technique attenuates base-line wander and high-frequency noise. Its fixed bandpass design may eliminate physiologically relevant information and inadvertently remove diagnostically important information and lacks adaptability to motion artifacts commonly present in ambulatory recordings. Median filtering, used in 14 studies, is particularly effective in baseline drift removal while preserving sharp QRS complexes and systolic peaks. However, it is less effective against high-frequency noise, and excessive smoothing may alter ST segments or distort PPG waveform morphology.

Eighteen studies have reported a notable trend of hybrid multi-stage denoising approaches. By combining techniques such as Butterworth, wavelet, and non-local means filtering [117], combination of methods offers improved robustness at the expense of greater computational complexity and extensive parameter tuning. Recently, adopted denoising methods based on autoencoders and GANs [121, 134, 139, 140, 156, 175] have shown extraordinary capability in overcoming complex motion artifacts. However, these approaches are data-hungry, computationally intensive, and susceptible to feature hallucination, raising concerns regarding the preservation of clinically meaningful signal characteristics.

3.5.2 Normalization

The z-score normalization is widely employed technique to reduce signal variability caused from physiological differences, acquisition artifacts, and device heterogeneity, being reported in 79 studies. By optimizing the data distribution, it facilitates cross-dataset consistency and improves the convergence of deep learning models. However, its sensitivity to outliers and its assumption of an approximately Gaussian distribution may distort the amplitudes of rare arrhythmias. Min-Max normalization is another commonly adopted approach, 34 studies. Owing to its ability to preserve the original waveform morphology, it is particularly suitable for applications in which signal shape carries diagnostic information. Nevertheless, the method is vulnerable to extreme values, as a single outlier can significantly influence the scaling of the entire signal.

3.5.3 Segmentation

Segmentation strategies have evolved considerably over time. A noticeable shift can be observed from algorithms that rely on accurate R-peak detection, such as the Pan–Tompkins method (9 articles), to fixed-size window-based approaches. Although the Pan–Tompkins algorithm works effectively in ideal conditions, its accuracy deteriorates in the presence of noise. Fixed-size windows, adopted in 54 studies, provide greater robustness to R-peak detection failures and are particularly suitable for end-to-end deep learning frameworks. However, these windows may include partially complete cardiac cycles, leading to inconsistencies in physiological content and potential loss of morphological information. R-peak-centered segmentation, adopted in 42 articles, conserves heartbeat morphology and is therefore advantageous for morphological analysis. Nevertheless, its performance remains highly dependent on accurate R-peak detection.

3.5.4 Data augmentation

Data augmentation plays a crucial role in overcoming severe class imbalance in arrhythmia datasets. Among the reviewed studies, SMOTE-based oversampling remained the most widely adopted approach (15 studies). Although SMOTE effectively balances minority classes and enhances the detection of in-frequent arrhythmias, the generated samples are not physiologically grounded and may have synthetic noise, thereby enhancing overfitting risk. Recent advances show, generative adversarial network (GAN)-based augmentation has emerged as an outstanding alternative and adopted in six studies. GANs can generate naturalistic synthetic signals and enhancing the representation of rare or noisy classes. However, instability to train the models and possibility that generated signals may not completely capture the diversity of clinical fluctuations remain core limitations. Transformation-based techniques, including noise injection and mix-up (11 studies), provide simple implementation and enhanced model robustness, but may generate patterns which are not real and that do not correspond to actual clinical scenarios.

To ensure reliability of augmented data, physiological consistency checks must be done. These may include assessing the distributions of morphological features, evaluating the preservation of rhythm characteristics, and testing model performance on independent real-world datasets. Additionally alternative strategies, including cost-sensitive learning, class weighting, and targeted acquisition of underrepresented classes, may offer more robust solutions while minimizing biases introduced by synthetic data generalization. Although aggressive denoising and image-based transformations can improve classification performance on benchmark datasets, they may do so at the cost of clinical interpretability. Conversely, the use of raw or minimally pre-processed signals preserve waveform morphology, thereby enhancing explainability and increasing greater physician confidence in prediction of models. Among the reviewed studies the exception was the framework proposed by Wang et al. [176], which employed deep active learning with skewness-based query strategies. Instead of generating synthetic samples, the method selectively acquired the most informative instances for annotation. Utilizing only 4,000 labeled samples corresponding to 14.3% of the training data, the approach achieved an accuracy of 99.34% for 14 heartbeat classes while reducing annotation requirements by approximately 85.7

DL model performance is substantially influenced by the specific preprocessing subgroup applied to the data, as demonstrated in Table 15.

As summarized in Table 15, the influence of pre-processing techniques employed on deep learning models is strongly dependent on the pre-processing subgroup applied to the data.

Although DL enables direct feature extraction from one-dimensional signals, changing signals into two-dimensional representations has been shown to improve the performance of CNN-based architectures. Among the available techniques continuous Wavelet Transform (CWT) is the most frequently adopted approach for generating scalograms, as it provides a balance between temporal and spectral resolution and thereby facilitating the detection of transient abnormalities such as ectopic beats [74, 94, 103, 147, 157, 158]. In contrast short-Time Fourier Transform (STFT) produces spectrograms, where the window size and overlap determines the trade-off between temporal and frequency resolution [147]. The Analytic Morse Wavelet offers adaptive time-frequency resolution and has demonstrated effectiveness for complex arrhythmias such

Table 15 Preprocessing methods used for DL model categories

DL Model Category	Most Common Preprocessing Methods	Typical Performance Range	Performance Insight
1D CNN	Z-score, Min-max, Fixed-length windows, R-peak centered, SMOTE	Acc: 96.79–99.68% F1: 0.92–0.9922	Highest when using R-peak + Z-score; excellent for beat-level tasks
Residual Networks	Wavelet, Hybrid Multi-Stage, Z-score, Min-max	Acc: 95.3–99.34% F1: 0.85–0.994	Wavelet + Hybrid gives best gains; very strong with ResNet family
2D / Image-based CNNs	Wavelet + CWT/STFT, Min-max, Image-based augmentation	Acc: 91–99.6% F1: 0.72–0.994	Wavelet + image transform is dominant; highest accuracy in this group
Tiny / Lightweight CNNs	Min-max, Light Butterworth, Fixed windows, No heavy denoising	Acc: 87.5–98.42% F1: 0.9217	Minimal preprocessing preferred; keeps model small and fast
Hybrid CNN-RNN	Wavelet, Hybrid denoising, Z-score, SMOTE, R-peak centered	Acc: 95.48–99.92% F1: 0.74–0.9988	Hybrid denoising + SMOTE + R-peak is the winning combination
Attention & Transformer	Fixed-length windows, Z-score, Overlapping, Wavelet	Acc: 98–99.84% F1: 0.89–0.9967	Fixed windows + Z-score is most common and highest performing
GAN / Autoencoder	GAN-based augmentation, Hybrid denoising, SMOTE	Acc: 97.41–99.6% F1: 0.96–0.996	GAN augmentation paired with GAN models gives the best results

as atrial fibrillation [74]. Similarly, the stationary wavelet-Hilbert transform integrates time-localized and frequency-domain information, whereas sophisticated approaches, including bispectral analysis combined with two-dimensional graph Fourier transforms, capture higher-order statistical analysis that are often overlooked by conventional methods [159, 177]. Furthermore image-generation techniques, such as Gramian Angular Fields, Recurrence Plots, and Markov Transition Fields, have also been successfully utilized to transform ECG signals into two-dimensional representations suitable for CNN-based arrhythmia classification [67, 98, 117].

3.6 RQ6: Which DL models are applied on ECG and PPG data for arrhythmia classification?

DL models for arrhythmia classification continue to evolve toward improved performance, generalizability, and clinical applicability, although simultaneously achieving all three objectives remains a major challenge. The reviewed studies indicate a clear evolution from traditional standalone CNN architectures to sophisticated hybrid, attention-based, and multimodal approaches, as shown in Fig. 5.

3.6.1 Spatial feature extraction models

CNNs are widely used for multi-lead analysis in ECG-based arrhythmia classification as they effectively learn spatial features from raw 1D signals and transformed 2D representations. Early shallow architectures, including enhanced LeNet [64] and 8-layer CNNs [66], achieved accuracies above 98%. Other notable approaches include Taguchi-optimized CNNs [108], wavelet-based CNNs [99], and residual architectures such as ResNet-based models [67, 74, 100, 117], many reporting accuracies close to 99%.

Recent research has focused on lightweight and hybrid CNN frameworks for wearable and edge applications. Miniature matched filter CNN (15 KB, 98.18% accuracy) [68], mobile-oriented CNNs (SqueezeNet, MobileNet) [103], a flamingo optimized CNN for IoT devices [114], and an ADCGNet, which is a dual channel CNN with Gabor filters and attention mechanisms [98], are used in resource-limited scenarios where speed and efficiency matter. LiteNet (454 params, 98.80%) [178] and PDNet (395 params, 98.20%) [179] could outperform other architectures of MIT BIH enabling device arrhythmia detection for wearable applications. CNNs have also been integrated into hybrid frameworks including CNN-LSTM/GRU [73, 94, 106, 109], CovHMM [71], and autoencoder-CNN-LSTM architectures [156], further improving sequential ECG analysis.

CNNs additionally perform effectively on transformed ECG representations, where wavelet- or CWT-based scalograms combined with ResNet, AlexNet, or multi-branch CNNs achieved accuracies between 96% and 99.34% [74, 104, 110, 113, 153, 158]. Novel concepts include the CNN without any preprocessing on the triad cycle image (99.16% accuracy) [161], lightweight CNNs [180] and hyperparameter-optimized CNNs (99.9% train accuracy) [160]. Advanced implementations include a dual-channel ResNet-ICBAM with 2D-CNN (97.8% accuracy) [153], a VGG16-based model with beat score maps (F1-score ≈ 0.93) [155], and multi-branch CNNs on spectrograms (99.34% accuracy) [158]. However, their limited capability in capturing long-range temporal dependencies remains a key limitation.

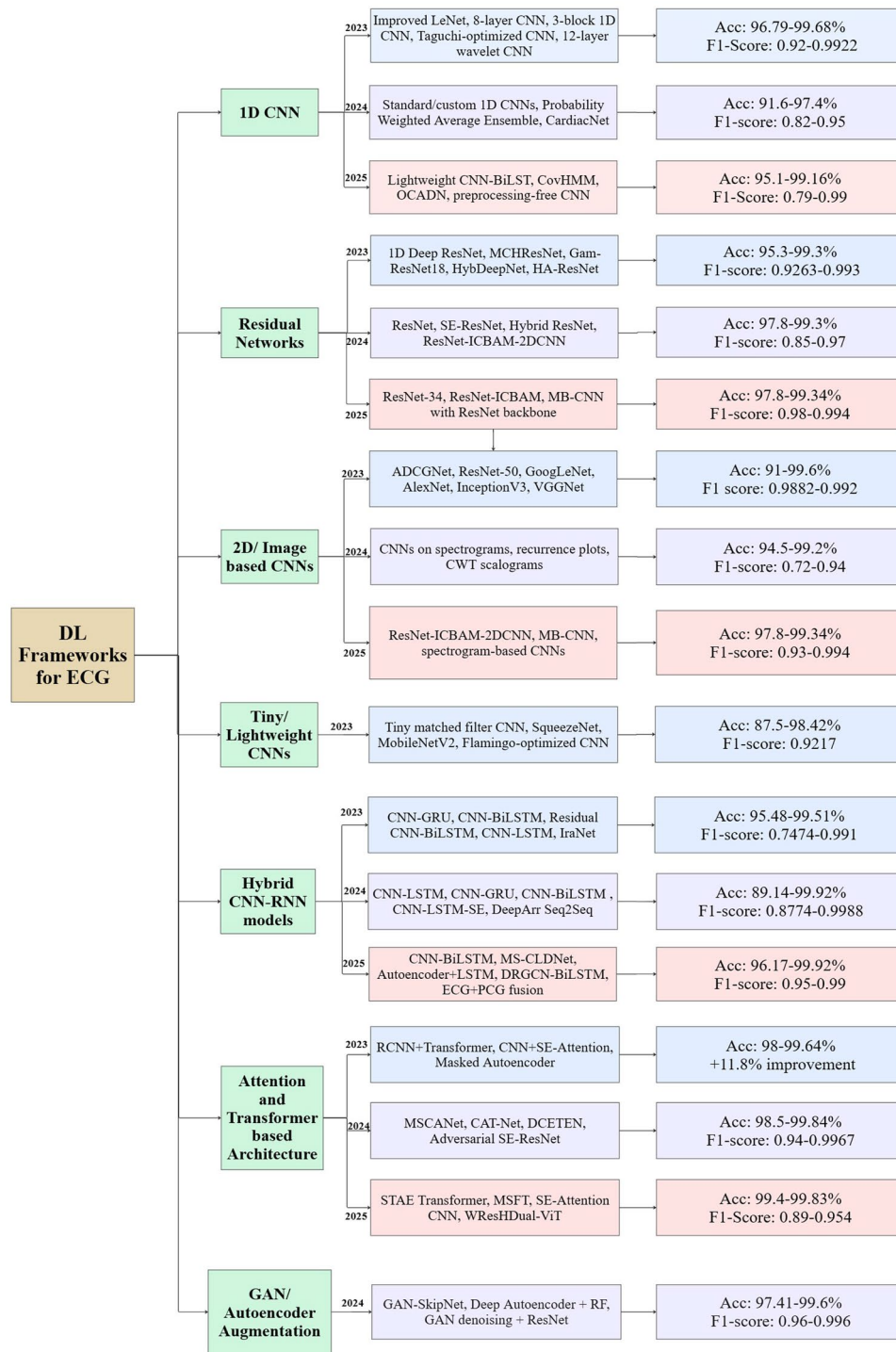


Fig. 5 Taxonomy of DL frameworks for ECG-based arrhythmia classification by architectural paradigm and year (2023–2025), with reported accuracy and F1-score ranges

3.6.2 Temporal modeling

Variants of RNN such as LSTM, BiLSTM, GRU, and BiGRU have been the most widely used models to extract temporal features from the data. In parallel, ResNet-based architectures have demonstrated strong performance and adaptability for arrhythmia classification. Early multi-channel and transfer learning-based ResNet models achieved

accuracies of 98.2–99.2% [67, 74, 104, 105], while attention-based HA-ResNet improved F1-scores to 99.3% [117]. Subsequent improvements included dual-channel fusion methods for improved interpretability [153], adversarial training techniques [138], and lightweight depthwise separable convolution models achieving 99.09% accuracy [145]. Ensemble methods also improved robustness against adversarial attacks [86, 101].

Hybrid CNN-RNN models leverage CNNs for spatial feature extraction and RNNs for temporal sequence modeling. Between 2023 and 2025, CNN-GRU and CNN-BiLSTM frameworks achieved accuracies ranging from 95% to 99.7% [66, 94, 106, 109, 118, 136]. Advanced ensemble and hybrid approaches, including CNN-LSTM-SE, ResNet-RNN, and Seq2Seq frameworks, further increased performance to nearly 99.9% [77, 126, 145, 146]. In 2025, lightweight and multiscale models such as compact CNN-BiLSTM and MS-CLDNet achieved >99% accuracy [33, 34], while multimodal fusion and explainable AI frameworks achieved robust accuracy in the 97%–99.92% range with clinical interpretability [37, 70, 151–154, 161, 163].

3.6.3 Attention mechanisms and transformers

Attention mechanisms and transformer-based models have become increasingly important in ECG arrhythmia classification by improving both discriminative feature learning and interpretability. Hybrid models such as CNN-Transformer-BiLSTM achieved 99.2% accuracy [65], while attention-based ensembles including CNN-SE reported accuracies up to 99.83–99.84% [75, 78, 91]. Residual models based on embedded attention Gam-ResNet18 and HA-ResNet provided 99.3% accuracy/F1-score [67, 117]. Self-supervised transformers further improved macro-F1 scores by over 11% [119], and adversarial attention mechanisms enhanced cross-database AF detection [138]. State-of-the-art transformer models such as STAE-Transformer and MSFT achieved up to 99.56% accuracy and macro-F1 scores of 99.4% with clinically validated interpretability [35, 36]. Dual-branch ViT-CNN models surpassed standard architectures on key datasets, obtaining macro-AUCs exceeding 90% [84]. Overall, transformer-based approaches consistently achieved accuracies above 99%, establishing them as powerful and interpretable solutions for arrhythmia classification.

3.6.4 Autoencoder and GAN-based approaches

Generative and autoencoder-based approaches have improved ECG arrhythmia classification by addressing noise, class imbalance, and computational constraints. A Generative Adversarial Network (GAN) based denoising approach achieved more than 30dB SNR improvement with a 99.1% F1-score on noisy ECG signals [61]. For imbalanced datasets, GAN-SkipNet combined with Bi-LSTM GAN generated synthetic minority-class beats, allowing the SkipCNN-LSTM classifier to reach a 99.6% F1-score [95]. Lightweight deep autoencoder models combined with Random Forest classifiers achieved 97.41% accuracy while substantially reducing latency by 70% and memory requirements by 90% [94]. Additionally, Wang et al. [181] developed a GAN-based beat generation model for 14 arrhythmia classes, achieving 99.28% accuracy on MIT BIH and reducing manual annotation time by nearly 89.43%.

3.6.5 Other techniques

Beyond conventional DL architectures, several unconventional approaches have demonstrated strong performance in ECG arrhythmia classification. Quantum and optimization-based frameworks, including Deep QNN with Exponential Political Optimizer [111] and FO-DCNN using Flamingo metaheuristics [114], achieved accuracies of 91.4% and 98.42%, respectively. Federated learning and domain adaptation frameworks, including FedECG and SF-ECG, enabled privacy-preserving cross-dataset classification with F1-scores up to 99.4% [69, 122]. Recent studies further explored graph-based learning [177], knowledge distillation for model compression [85, 144], reservoir computing with 92.09% AUC [137], and LLM-guided feature engineering achieving 96.2% accuracy [148]. Recent architectures such as Bi-CapsNet [149], FOCAE autoencoders [83], interpretable KANs [76], and fusion-based CNN-GCN and HMFF models achieved accuracies between 97% and 99.3% [37, 151, 152].

The architectural progression of models from CNNs to CNN-RNN hybrids to Transformers represents diminishing performance gains, despite rapidly increasing computational cost. Although transformer-based models achieved the highest gains in meta-regression analysis (+0.845% over baseline 1D-CNNs [35, 36]), these improvements required 10–1000× greater computational resources (Fig. 9). Pareto frontier analysis suggests that beyond 1 M parameters and 1 GFLOPs, further accuracy gains below 0.5% require exponentially higher computation. Consequently, many large multimodal fusion models (mean 25.4M parameters) remain outside the Pareto-optimal frontier despite reporting accuracies above 98% [84, 85].

3.6.6 PPG-based classification

Early PPG arrhythmia research used deep CNNs adapted from VGGNet for multi-class classification. A 1D-CNN trained on more than 118k paired PPG–ECG samples achieved 85% accuracy with an AUC of 0.978, representing one of the first end-to-end DL frameworks for PPG arrhythmia classification [52]. However, classification performance for PAC/PVC dropped, and model generalization to wearable environments was not assessed. Concurrently, feature-engineering approaches using handcrafted morphological descriptors with SVM, RF, and DT classifiers achieved retrospective accuracies close to 98%, but performance dropped to nearly 94% on prospective ambulatory datasets, revealing poor cross-setting generalization [27]. Recent developments have increasingly adopted hybrid CNN-Transformer architectures (Fig. 6), including channel-attention fusion frameworks for clinical ablation PPG data that achieved F1-scores near 86.8%, emphasizing the importance of integrating local and global feature representations [169].

Res-BiANet, integrating ResNet, BiLSTM, and multi-head attention, demonstrated strong spatial–temporal modeling capability with approximately 92% accuracy and 86.9% F1-score [170]. Bi-GRU frameworks applied to smartwatch-derived multimodal data integrating PPG, HR, and accelerometer signals achieved 97.3% AF detection accuracy with robustness to motion artifacts, although performance for PAC/PVC remained lower [174]. Similarly, motion artifact reduction using 1D-CycleGAN improved AF detection by nearly 5%, allowing the Self-AFNet framework to achieve ECG-comparable performance of 98–99% [175]. Other hybrid models, including DenseNet-ABiLSTM, addressed spatial and temporal dependencies interpretably but achieved comparatively

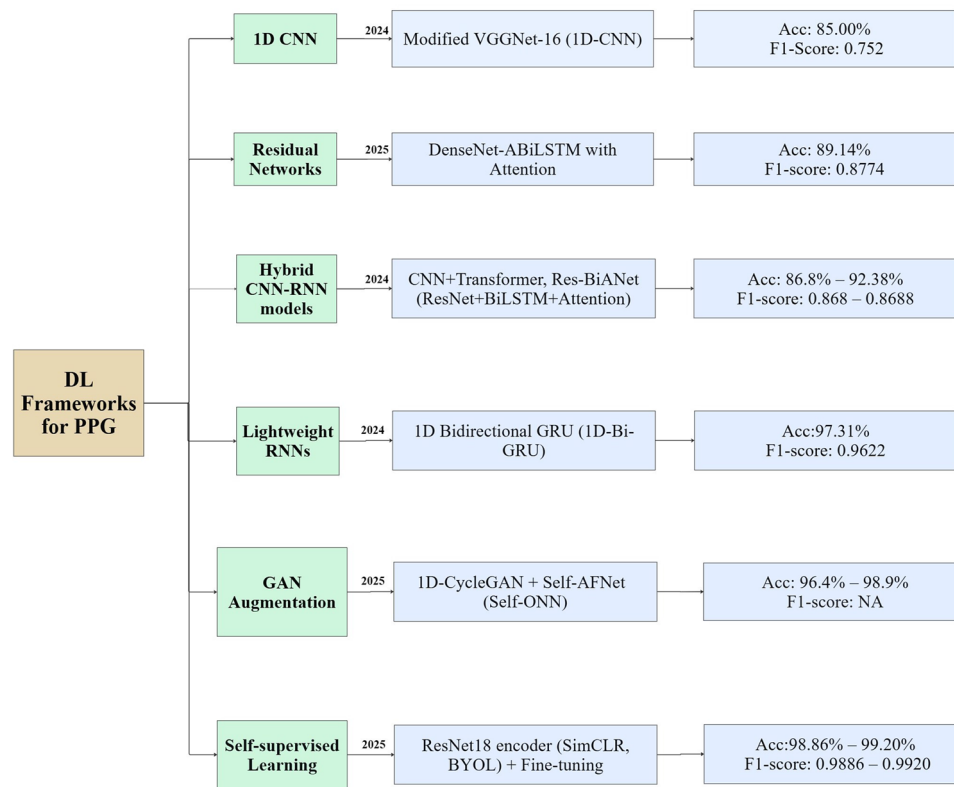


Fig. 6 Taxonomy of DL frameworks for PPG-based arrhythmia classification by architectural paradigm (2023–2025), with reported accuracy and F1-score ranges

moderate performance ($\sim 89\%$ accuracy, $F1 \sim 87.7\%$) [31]. More recently, contrastive learning using ResNet encoders achieved substantial improvements in AF detection, increasing F1-scores by 35% over supervised baselines using minimal labeled data [15]. Additional approaches such as TQCPat [173] and EP study-annotated multi-scale ResNets [14] reported strong accuracies of approximately 91–99%, although evaluation remained limited to smaller datasets.

Despite these advances, PPG-based DL systems generally underperform compared to ECG-based approaches. Although ECG-based DL systems attain classification accuracy greater than 95% on benchmark standard databases [75, 85], PPG-based models yield performance in the range of 80–90%, and are primarily limited to binary AF detection tasks [26, 167]. This disparity originates from the indirect physiological nature of PPG signals, which measure peripheral blood volume changes rather than direct cardiac electrical activity [162]. As a result, arrhythmias with similar hemodynamic profiles, such as PAC and PVC, are difficult to distinguish using PPG alone [170, 172]. To overcome these limitations, recent studies increasingly explored multimodal ECG-PPG fusion strategies for improved classification performance and clinical applicability [34, 167].

To consolidate the architectural survey presented in this section, Table 16 and Table 17 summarize representative ECG- and PPG-based studies, respectively, spanning dataset, architecture, reported accuracy/F1 score, computational complexity, advantages, limitations, and clinical application scenario. FLOPs were reported by fewer than 8 of the 119 reviewed studies (Sect. 3.6.9). Also parameter count and inference latency were used as complexity proxies where FLOPs are unavailable. The tables together show that no single

Table 16 Comprehensive comparison of representative DL architectures for ECG-based arrhythmia classification

Model	Dataset	Architecture	Acc / F1	Complexity	Advantages	Limitations	Application
Tiny MFCNN [78]	MIT-BIH	Lightweight CNN	98.18%	15 KB	Ultra-low footprint	Low minority-class sensitivity	Edge / MCU monitoring
CNN-BiLSTM Ensemble [77]	MIT-BIH	Hybrid CNN-RNN	99.88%/ 99.3%	NR	Best MIT-BIH result	Intra-patient split bias	Offline validation
1D-CNN + Attention [92]	PTB / PTB-XL	Attention CNN	99.83%/ 99.48%	0.89 ms	Sub-ms inference	GPU-only latency	Clinical 12-lead screening
CASLCNet [79]	CPSC 2018	CNN, Lipschitz-constrained	91.03%/ 91.22%	NR	Adversarial robustness	Harder 9-class task	Rhythm screening
xECG-Beats [80]	INCART	Transfer learning ResNet	98%/ –	VGG/ResNet backbone (transfer learning; size NR)	Mitigates small cohort	Non-standard sampling rate	Cross-dataset validation
CNN-BiLSTM [81]	PhysioNet/ CinC	Hybrid CNN-RNN	98.5%/ 98.4%	NR	Spatial + temporal fusion	Short, undisclosed test set	Single-lead screening
CNN-BiLSTM+Attn [82]	Chapman-Shaoxing	Hybrid + Attention	98.8%/ –	NR	Improves discrimination	Misses paroxysmal events	12-lead rhythm classification
FOCAE [83]	Real wearable ECG	Fuzzy autoencoder	96.8%/ 96.5%	NR	Validated on real noisy data	Lower vs. curated benchmarks	Wearable deployment
LiteNet [182]	MIT-BIH	Ultralightweight CNN	98.80%	454 params	Smallest model, strong acc.	Weak minority-class recall	Implantable devices
PDNet [183]	MIT-BIH	Ultralightweight CNN	98.20%	395 params	Comparable to LiteNet	Same recall concern	Low-power wearables
FedECG [69]	Multi-institution	Federated ResNet	F1 99.4%	ResNet-9 backbone (federated); client count NR	Distributed, privacy-preserving	Non-IID convergence issues	Multi-hospital training
STAE-Transformer [35]	PTB/PTB-XL	Transformer + VAE	99.56%, F1 99.4%	NR	Clinician-validated attention	High compute cost	Explainable screening
MSFT [36]	CPSC 2018	CNN + Transformer	F1 99.4%	NR	Multi-scale fusion	Higher FLOPs vs. 1D-CNN	Multi-class screening
WResHDual-ViT [84]	PTB-XL	Dual-branch ViT-CNN	AUC > 90%	NR	Strong large-scale acc.	Not Pareto-optimal	Research benchmarking
MTSD [85]	PTB-XL	Multi-teacher distillation	F1 > 0.74, AUC > 0.92	NR (compression via distillation; exact param. count not disclosed in accessible text)	Efficient via distillation	Added training complexity	Lightweight deployment
CovHMM [71]	MIT-BIH	CNN-HMM hybrid	> 98%	61K params	Efficient hybrid design	Less interpretable	Embedded detection
DRGCN-BiLSTM [70]	MIT-BIH	Graph-hybrid CNN-RNN	99.92%/ 99.33%	0.63 M params	Best Pareto accuracy	Higher inference cost	Clinical diagnostics

Table 16 (continued)

Model	Dataset	Architecture	Acc / F1	Complexity	Advantages	Limitations	Application
GAN-SkipNet [95]	MIT-BIH (aug.)	GAN + BiLSTM	F1 99.6%	NR	Fixes class imbalance	GAN training instability	Rare-class augmentation
HA-ResNet [117]	2D signal	Residual + Attention	F1 99.3%\$	NR	Improves 2D interpretability	Needs 2D transform	Diagnostic screening
GanResNet18 [67]	Image feature fusion	2D CNN + Attention	99.3%\$	ResNet-18 backbone (transfer learning; size NR)	Attention on fused images	Preprocessing overhead	Diagnostic screening

Table 17 Comprehensive comparison of representative DL architectures for PPG-based arrhythmia classification

Model	Dataset	Architecture	Acc / F1	Complexity	Advantages	Limitations	Application
1D-CNN (VGGNet) [52]	Fuwai Hospital, > 118k pairs	Spatial CNN	85.0% AUC 0.978	NR	First end-to-end multiclass PPG DL	Weak PAC/PVC discrimination	Clinical monitor feasibility
CNN + Transformer [169]	Liu et al. PPG	CNN + Transformer, channel attn.	F1 86.8%	NR	Local + global feature fusion	No accuracy reported	Ablation-lab PPG analysis
Res-BiANet [170]	Liu et al. PPG	ResNet + BiLSTM + Attention	92.38%/ 86.88%	NR	Best on this benchmark	PAC/PVC still hard	Smartwatch multiclass screening
KNN [171]	PhysioNet/CinC 2015	Classical ML	97.30%	41 features (PCA-reduced)	Simple, competitive baseline	Poor OOD generalization	ICU bedside monitoring
SVM+RBF + ABC-PSO [172]	CapnoBase	Metaheuristic SVM	95.12%/ 95.00%	41 subj., 29,520 segments (ABC-PSO reduced)	Competitive on small cohort	Small cohort (42 subj.)	Small-cohort validation
TOCPat + SVM/KNN [173]	Liu et al. PPG	Quantum-inspired features	91.30%/ 85.55%	46,827 signals, 10-fold CV	Interpretable pipeline	No external validation	PPG feature-engineering baseline
1D Bi-GRU [174]	Pulsewatch trial	Multimodal temporal	97.31% (AF)	NR	Robust to motion artifacts	AF-only scope	Smartwatch AF screening
Self-AFNet [175]	Wrist PPG	CycleGAN + classifier	98–99%	NR	Closes ECG–PPG gap	GAN-based, compute-heavy	Wearable AF under motion
DenseNet-ABiLSTM [31]	PPG (multiclass)	Hybrid DenseNet + BiLSTM	~89%/ ~87.7%	NR	Interpretable decomposition	Moderate performance	Multiclass PPG (research)
Contrastive ResNet18 [15]	MIMIC-III	Self-supervised (SimCLR/ BYOL)	F1 99.20%	NR	Reduces label need	ICU noise/bias	Label-scarce AF detection
Multi-scale ResNet [14]	Clinical PPG (EP-annotated)	Multi-scale ResNet	91–99%	NR	High-quality EP annotation	Small dataset scale	PPG morphology research

paradigm dominates across all axes. Attention/transformer and hybrid CNN-RNN models achieve the highest accuracies, ultra-lightweight CNNs (e.g., LiteNet, PDNet) offer the smallest computational footprint, and generative and federated approaches address imbalance and privacy constraints at the cost of added training complexity.

3.6.7 Loss functions and optimizers

The Adam optimizer is the predominant optimization algorithm used in DL models for ECG and PPG analysis (Fig. 7). Its effectiveness arises from adaptive learning rates that integrate the benefits of AdaGrad and RMSProp, enabling efficient optimization on noisy and sparse physiological signals. Adam is computationally efficient, robust to noisy physiological signals, and typically achieves strong performance with limited manual tuning. However, it may converge to suboptimal minima and can be outperformed by carefully tuned SGD-based optimization schedules. Moreover, Adam requires greater memory resources than conventional optimization algorithms.

Less frequently used variants of Adam include AdamW and Adamax. AdamW, which decouples weight decay from gradient updates, improves regularization and often yields better generalization and performance [84, 97]. However, it is relatively more complex and less standardized than conventional Adam. An infinity norm-based variant, Adamax is stable in certain architectures [78, 79, 158], although its practical performance is often comparable or inferior to Adam.

Traditional SGD and SGDM remain important alternatives because of their simplicity, interpretability, and ability to converge to flatter minima associated with improved robustness and generalization. However, SGD-based methods require extensive manual tuning of learning rate schedules and momentum parameters, exhibit slower convergence, and are more sensitive to saddle points than adaptive optimizers.

Historically, effective optimizer in training RNN/LSTM architectures that performs good for non-stationary objectives is RMSprop [94, 98]. However, absence of bias correction and momentum mechanisms has limited its adoption in modern DL applications, where Adam is generally preferred.

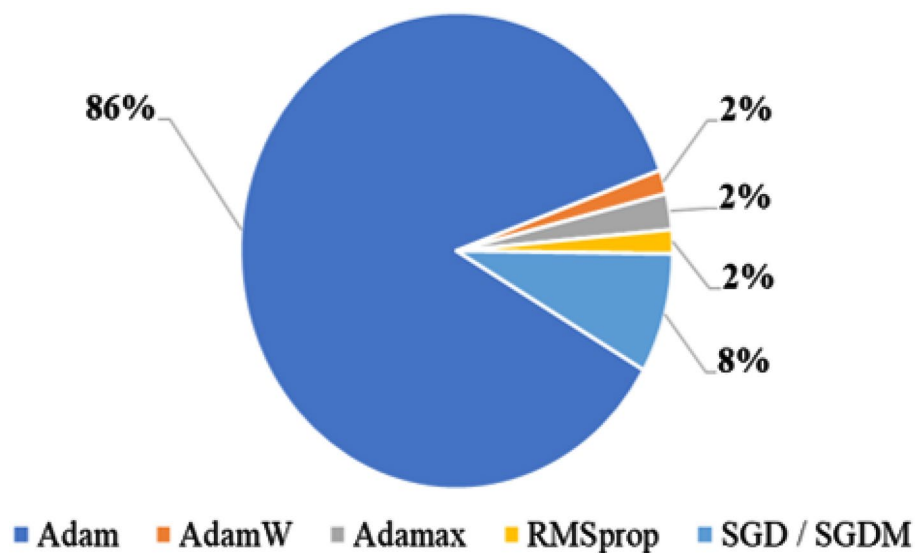


Fig. 7 Distribution of optimizers used in reviewed studies

Loss function optimization has become increasingly important in handling dataset imbalance for arrhythmia classification. Categorical Cross-Entropy remained the standard loss for multi-class classification but performed poorly on imbalanced datasets (Fig. 8). Focal Loss improved minority-class learning by emphasizing difficult samples [27, 35, 36, 38, 66, 125, 161], while Weighted Cross-Entropy offered a simpler balancing strategy based on class weights [37, 72, 118, 120]. For specialized tasks, Binary Cross-Entropy was the standard for binary classification [60, 133, 163], and Multi-Label Cross-Entropy managed co-existing diagnoses [84, 87, 97, 101, 120]. Other task-specific losses included Hybrid Loss (Focal + Dice) [35], Hinge Loss [141], MSE for reconstruction and pretraining [88, 119], and adversarial losses such as LSGAN for data synthesis and domain adaptation [61, 134, 138]. A recent contribution by Zhang et al. [184] introduced the Class-wise Statistics Loss (CWS), improving F1-score by 8.5–21.4% and balanced accuracy by up to 27.8%.

3.6.8 Hyperparameter optimization

Hyperparameter selection remains one of the most challenging practical aspects of DL model development. Despite the rise in using automated optimization techniques, the majority of papers use arbitrarily chosen or predefined hyperparameters, which hinders replication and comparison across studies. The hyperparameters that influence convergence and generalization include learning rate, batch size, number of hidden layers, and regularization. Moreover, overfitting might occur due to extensive optimization on validation sets, leading to inflated accuracy reports. Hence, the reporting of the methods used for searching and validation is imperative for the maturity of this research area.

Developments in hyperparameter optimization demonstrate an increasing trend towards automation. Nature-inspired metaheuristics found wide application in 2023, namely Grey Wolf Optimization [63], Farmland Fertility optimization [94], Flamingo Optimization [114] and Grasshopper Optimization [124]. While such methods successfully explore non-convex spaces, they are expensive in terms of computational resources and are extremely hardware dependent. In 2024, popularity of Bayesian Optimization increased due to high sample efficiency achieved through the building of probabilistic model of the objective function [125, 127] (in several papers, Bayesian optimization was used together with grid or random search in order to explore the space of parameters at

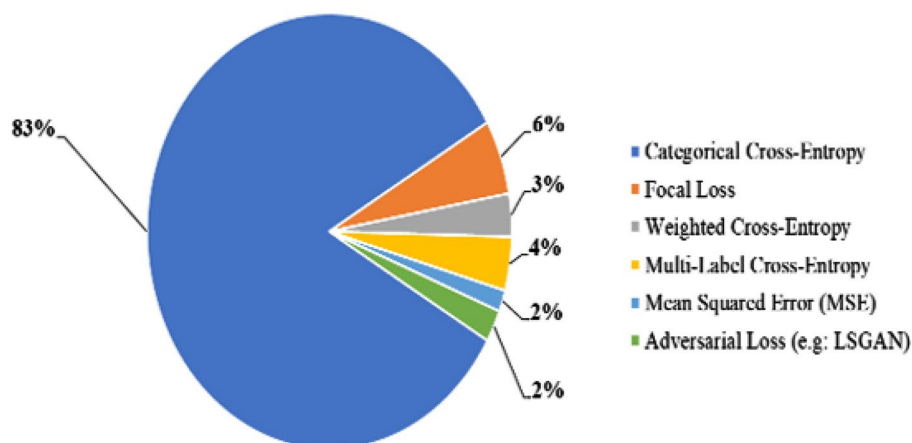


Fig. 8 Distribution of Loss functions used in reviewed studies

first) [129, 160]. In 2025, reinforcement learning methods were developed for hyperparameter and architecture optimization during the training process [38]. Due to federated learning approach, non-IID distribution of the data became another challenge for optimization procedure [162]. Overall, these developments reflect a broader transition toward automated optimization pipelines for clinically deployable arrhythmia classification systems.

3.6.9 Performance metrics

Performance parameters that are commonly reported for DL-based systems are accuracy, precision, sensitivity, specificity, and F1 score. However, the clinical significance of all these measures relies on the potential adverse effects of wrong classifications. False negatives of serious types of arrhythmias, like VT and VF, will mean delays in providing the necessary therapy and fatal consequences, while false positives of harmless types of arrhythmia will just mean additional burden of medical treatment and extra stress. Absence of standardized and statistically robust evaluation procedures and criteria is one of the biggest problems with ECG and PPG arrhythmia classification studies.

While many articles adhere to AAMI EC57 specifications, there are also those that use customized or binary class specifications, thereby making direct comparisons impossible. Another major problem with many of the studies involving PPGs is that there is often inadequate information about their methodology, thus lowering their reproducibility. Moreover, while accuracy is the primary focus in most articles, sensitivity in terms of minority classes is of greater clinical relevance. To improve reliability, future studies should report independent test-set performance, class-wise metrics, confidence intervals, variance across folds, and standardized class definitions.

Direct comparison of performance metrics across studies is often misleading because of differences in evaluation schemes, dataset granularity, and class distributions. Intra-patient splits (72% of beat-level MIT-BIH papers) yield consistently better accuracy than inter-patient splits. Five-class beat-level datasets (MIT-BIH) are more challenging than 9-class rhythm-level (CPSC 2018) and multi-label record-level (PTB-XL) datasets. The Normal class of MIT BIH and PTB-XL are 87% and 52% respectively. Consequently, metrics such as macro-F1 cannot be directly compared across datasets. Detailed comparisons are provided in Table 12 and Sect. 4.1. Hence, future studies should explicitly report evaluation protocols, dataset granularity, and F1-score aggregation strategies alongside performance metrics.

Another major gap is insufficient reporting of computational efficiency of the DL model. Only a small proportion of studies report parameter count (14 studies), FLOPs (8 studies), or inference latency (5 studies). Model complexity varies widely, ranging from ultra-lightweight architectures such as MSCANet (34,000 parameters, 8 ms) and Tiny MF-CNN (15KB) [68] to large multimodal fusion models averaging 25.4 M parameters [84, 85]. Current benchmarking practices encourage increasing model complexity for marginal performance gains beyond 98% accuracy, despite limited additional clinical utility. Lightweight models with fewer than 100,000 parameters can still achieve accuracies above 94%, although minority-class F1-scores often decline by 10–15% points [68, 144]. Approaches such as knowledge distillation [85, 144], federated learning [69, 162], and efficient attention mechanisms [91] could yield interesting and untapped

opportunities. Future work should routinely report parameter count, FLOPs, inference speed, and memory requirements alongside predictive performance [68, 85, 144, 162].

3.7 RQ7: How well do DL models generalize across different datasets and recording conditions?

External validation remains limited in current ECG- and PPG-based arrhythmia research, with only 22 of 119 studies (18.5%) evaluating models across datasets. Models trained predominantly on MIT-BIH experienced performance drops of 6–14% in accuracy and 8–18% in F1-score when evaluated on external datasets including INCART, PTB/PTB-XL, Chapman-Shaoxing, CUIDB, and wearable PPG datasets. Minority arrhythmia classes and rhythm-level tasks exhibited the greatest degradation. In addition, most studies failed to report confidence intervals, fold-wise variance, or statistical significance, making many reported improvements difficult to validate.

The primary causes of poor generalization include variations in sampling rates, lead configurations, acquisition settings, and limited demographic diversity. In contrast, stronger cross-dataset robustness was achieved using domain-adversarial learning, federated learning frameworks such as FedECG [69], attention mechanisms, batch normalization with dropout, and multi-dataset pretraining, reducing performance degradation to $\leq 5\%$. Ensemble approaches combining dataset-specific models further improved robustness by an additional 3–6%.

These findings highlight the need for standardized and clinically realistic cross-dataset evaluation protocols. Future research should evaluate models on multiple external datasets using inter-patient validation and demographic stratification. Reporting standards should include per-class F1-score, accuracy degradation, and 95% confidence intervals for each transfer experiment. Moreover, systematic ablation studies on normalization, attention mechanisms, augmentation, and multi-task learning should accompany evaluations of ensemble strategies such as voting and stacking. Such protocols are essential for transitioning from benchmark-specific optimization toward clinically robust and generalizable arrhythmia classification systems.

3.8 RQ8: What challenges affect the clinical translation of DL-based arrhythmia classification systems?

Translation of DL-based arrhythmia classification from benchmark-oriented research into clinical applications represents one of the biggest and most important challenges in the field. As shown in the meta-analysis (Sect. 4.1), a number of approaches already achieve above 99% accuracy on existing datasets, indicating that the problem is not in the performance of the algorithms. The major difficulties are complex and include: (i) the regulatory challenge that requires clinical trial data instead of retrospective benchmarking, (ii) the need for explanations and interpretations to allow clinicians to understand and adopt the predictions, (iii) operational challenges such as computational latency, battery efficiency and heterogeneity of patients, especially in wearable devices, and (iv) healthcare system-level challenges such as governance, legal liability and reimbursement issues. Notably, less than 5% out of 119 analyzed studies were focused on these translation-related challenges.

3.8.1 Regulatory and validation requirements

The regulatory framework for DL-based arrhythmia classification depends on the intended diagnostic application and associated clinical risk. According to the Food and Drug Administration (FDA) of United States, AI/ML-based cardiac diagnostic software is classified as Software as a Medical Device (SaMD) under Part 880 of 21 CFR [185]. Existing-device-equivalent devices may follow the 510(k) pathway with requirements for analytical and limited clinical validation, whereas novel architectures generally require the De Novo process involving formal risk assessment, IEC 62,304-compliant software documentation, and clinical evaluation studies. Fully autonomous diagnostic systems require Premarket Approval (PMA) supported by randomized clinical trials. FDA recommendations further specify sensitivity $\geq 90\%$ for critical arrhythmias such as VT/VF and PPV $\geq 75\%$.

The European Medical Device Regulation (EU MDR) 2017/745 Article 61 [186] similarly mandates clinical evaluation supported by clinical investigations rather than literature equivalence alone, especially for adaptive AI/ML systems. Article 61(4) additionally mandates a Predetermined Change Control Plan (PCCP) specifying permissible post-deployment model modifications. In India, the Central Drugs Standard Control Organization (CDSCO) Medical Device Rules 2017 [187] classify cardiac diagnostic software as Class B or C devices, with VT/VF detection systems generally falling under Class C because of higher clinical investigations and compliance with IEC 62,304 and IEC 62,366 standards.

Prospective testing is necessary for regulation of algorithms since retroactive evaluation cannot prove that the device operates safely under pathological, demographic, and signal conditions other than those used during training. For an arrhythmia classification software as a medical device, a prospective validation must include: (i) pre-registered clinical protocols, (ii) prospective consecutive patient enrollment to minimize spectrum bias, (iii) demographic subgroup analysis across age, sex, ethnicity, and BMI, and (iv) gold-standard annotation by multiple cardiologists with reported inter-rater agreement. FDA power calculations suggest that more than 2,000 patients may be required to achieve statistically reliable validation for low-prevalence arrhythmias.

Regulatory agencies also increasingly emphasize post-market surveillance. The FDA PCCP framework [188] allows predefined model updates without full resubmission if retraining procedures, data sources, rollback plans, and performance monitoring are documented. Similarly, under the EU MDR's Post-Market Clinical Follow-up (PMCF), performance data are to be collected systematically along with annual safety reports. These considerations highlight the need to integrate regulatory compliance into DL model development from the outset.

3.8.2 Real-world data challenges

In practical deployment settings, physiological signals are frequently affected by noise and variability that are insufficiently represented in benchmark datasets. Motion artifacts during ambulatory monitoring, sensor placement variability, and inter-device differences can significantly degrade signal quality and model performance [27, 47]. In PPG-based systems, additional factors such as skin tone, ambient light conditions, and peripheral perfusion influence signal fidelity and can introduce bias [14]. Furthermore,

variations in ECG morphology due to patient-specific factors, comorbidities, and medication effects can reduce model generalizability across populations [52].

Wearable arrhythmia monitoring systems must satisfy strict real-time and energy requirements. Inference latency must remain below the inter-beat interval (approximately 600–1000 ms at resting heart rates) to avoid missed arrhythmic events, while end-to-end processing latency should remain below 200 ms for timely clinical alarms. Energy constraints are equally critical, with average inference power typically required to remain below 1 mW for multi-day wearable operation [68, 144]. Although bedside monitors may tolerate delays up to 2 s, false alarm rates must remain below one per 8 hours. Importantly, the aforementioned latency values (0.179–0.89 ms) have been reported using a GPU, not ARM Cortex-M or RISC-V microcontroller present in the commercially available wearable devices. Given GPU-to-MCU performance differences of 50–200 $\times$, actual deployment latency may approach or exceed the 200 ms clinical threshold.

Only a small fraction of studies (2.5%) reported inference energy consumption. Commercial cardiac patches run 14 days using ≈ 180 mAh at 3.7 V (total 670 J). In case of baseline power of 10 mW for the MCU, the average inference needs to be less than 1 mW. Hence, it should use ≈ 1 μ J for each of 1,000 inferences per hour. Large architectures such as ResNet-50 and transformer models (> 25 M parameters) consume 100–1000 $\times$ more energy than lightweight models (< 100 K parameters). INT8 quantization decreases energy by 4 to 16 times and degrades accuracy by 0.3 to 1.5% on average [85, 144] and 3 to 7% F1 degradation. Notably, no reviewed study reported class-wise effects of quantization.

As identified by the Pareto analysis in this review, only two neural network models managed to satisfy both sub-millisecond inference latency and more than 99% accuracy requirements (CNN-BiLSTM: 0.179 ms; 1D-CNN+Attention: 0.89 ms), and sub-100K parametric models (MSCANet: 34K; CovHMM: 61K) scored over 94% accuracy under MCU constraints. Therefore, future studies targeting embedded deployment should systematically report inference latency, energy consumption, memory footprint, and quantization-induced degradation on target hardware. Techniques such as quantization-aware training, pruning, and neural architecture search remain promising approaches for achieving clinically deployable real-time systems [85, 144].

Performance is additionally affected by demographic diversity and acquisition variability. Age-related ECG morphology changes, medication effects, pacemaker interference, electrode placement variability, and skin-tone-related PPG signal degradation all influence performance. PPG is impacted by skin color (melanin concentration reduces SNR up to 6 dB for Fitzpatrick V–VI) [14], lighting conditions, and body area. Nevertheless, demographic factors such as skin tone remain largely unreported in existing datasets.

3.8.3 Clinical workflow integration and physician acceptance

The DL models are anticipated to support the clinicians in decision making and not serve as a replacement [45]. Nevertheless, limited interpretability remains a significant obstacle to clinical adoption, contributing to reduced physician trust, alarm fatigue, workflow interruption, and poor usability. Output should include probabilities, confidence indicators, and regions of signals that caused decision-making.

For clinicians to accept, models must be able to provide verifiable reasoning. Black-box binary outputs are insufficient for medical liability and regulatory requirements outlined by the FDA [185]. Three explainable aspects are needed: signal level attribution (what parts of the waveform influenced the result), feature level attribution (what features influenced the result), and uncertainty quantification (confidence/deferred). Although Grad-CAM and attention-based explanations are commonly used, fewer than 10% of studies validated attention maps against clinically meaningful ECG structures [35, 126, 156]. Consequently, many attention maps may reflect dataset artifacts rather than true physiological reasoning.

Commonly used post-hoc techniques include Gradient-weighted Class Activation Mapping (Grad-CAM), SHapley Additive exPlanations (SHAP), and Local Interpretable Model-Agnostic Explanations (LIME). Grad-CAM identifies discriminative temporal regions but may emphasize noise or baseline drift instead of relevant cardiac morphology [126, 156]. The “artifact-driven attribution” cannot be detected by accuracy measures only and needs cardiologist ground truth data validation, and only less than 10% of transformer models use it [35]. SHAP offers attribute generation at a theoretically guaranteed level, yet its computational complexity is $O(2^n)$, making it difficult to use in practice due to the prohibitive time cost. LIME offers local interpretability but can generate physiologically implausible perturbations.

SHAP has shown that RR-intervals’ contribution to PPG beats morphological features’ by a quantifiable measure [76, 106]. In contrast, attention-based architectures such as the STAE-Transformer [35] validated attention maps using cardiologist annotations, while interpretable architectures such as KANs [76] introduced symbolic and clinically inspectable functions. In terms of interpretability for PPG, little progress has been made since no studies have produced attention maps validated clinically.

DL classifiers also exhibit several critical failure modes, including patient-specific morphological overfitting, artifact-driven predictions, minority-class suppression, and highly confident out-of-distribution errors. Methods to quantify uncertainty, such as Monte Carlo dropout [90], deep ensembles, and conformal prediction, may discover failure modes and by passing uncertain cases to clinicians.

Integration of Electronic Health Records (EHR) demands model outputs compatible with the HL7 FHIR, SNOMED-CT, and DICOM standards. Alarm fatigue—up to 99% of cardiac monitor alarms can be considered non-relevant [188]—is the main cause of alarm non-compliance. Classifier with specificity of 95% for arrhythmia prevalence of 5% produces PPV that is just 50% (half of the time, the alarms are irrelevant), which inevitably causes system non-utilization. In order for a model to provide clinically acceptable PPV (>75–80%), its specificity must be greater than 98%. Only 12% of the reviewed works satisfied such criteria.

3.8.4 Ethical, legal, and economic considerations

Beyond technological advancement, several non-technical factors significantly contribute the clinical implementation of AI-based arrhythmia classification systems. These include liability, concerns regarding privacy, security, regulatory requirements and others. Compliance to the Digital Personal Data Protection Act (DPDP Act, 2023) in the context of India is crucial to manage patient data securely [189]. Reimbursement, cost-effectiveness, and infrastructure are other areas that play a key role in making a

decision about implementing these technologies. To support equitable healthcare delivery fairness should be ensured and bias should be minimized among diverse demographic groups.

Successful clinical translation will therefore require comprehensive evaluation frameworks incorporating cross-center and cross-dataset validation using diverse patient populations. Combining explainable AI with lightweight and efficient architectures may further enhance trustworthiness, interpretability, and practical deployment in healthcare environments.

3.8.5 DL approaches for arrhythmia localization

Unlike most of the existing DL algorithms developed for arrhythmia classification, which mainly emphasize on temporal classification of different arrhythmia types, localization of the source region in anatomy plays a much more important role in catheter ablation treatment clinically. Precise localization will help to guide the intervention procedure effectively and avoid the necessity of complicated invasiveness procedures. Several latest DL frameworks, especially those based on hybrid recurrent and convolutional deep models with attention techniques, have demonstrated superior results in localizing IVA foci origins or other abnormal foci regions from the ECG signal data. Recurrent convolutional neural network (RCNN) with attention techniques by Shahsavari et al. [29] (2024) used 12-lead ECGs to detect whether the origin was right or left ventricle outflow tract. The model adopted the CNN-based spatial fusion among channels, the bidirectional LSTM for sequence modeling, and an attention mechanism for explainable modeling. This model achieved 94.3% accuracy and 0.933 AUC based on a dataset consisting of 334 subjects confirmed by the ablation study [29]. Other relevant studies include PVC and AF driver localization [190, 191]. Unlike type classification, which can work with single-lead or PPG signals, origin localization requires multi-lead ECG data to capture spatial morphology differences and demands precise anatomical ground truth. Hybrid architectures effectively model spatiotemporal patterns, but challenges remain in dataset imbalance, generalizability, and limited spatial resolution.

4 Discussion

Based on the methodological contributions reviewed in the literature and the observations presented in Sect. 3, this section provides a comparative analysis of existing approaches, meta analysis, highlights major research gaps, and discusses future directions in DL-based arrhythmia classification.

4.1 Meta-Analysis

In order to quantitatively estimate pooled performance, assess heterogeneity, and identify methodological factors influencing outcomes Meta analysis was performed. Random-effects pooling was performed to obtain consolidated accuracy estimates while incorporating variance in between studies. Subgroup analyses on MIT BIH were conducted to compare performance across different model architectures, signal modalities, and preprocessing strategies. Further, to evaluate the independent influence of methodological factors on classification accuracy, Weighted least-squares (WLS) meta-regression was applied. Pareto frontier analysis was also employed to evaluate the balance

between model complexity and predictive performance. Funnel plots were used to examine potential publication bias.

Pareto frontier analysis of the accuracy–FLOPs trade-off identified two architecturally distinct optimal models, as depicted in Fig. 9. MSCANet (Spatial, 99.35% accuracy, 0.034M parameters, 0.15 GFLOPs, 8ms inference) represents the efficient frontier, achieving near state-of-the-art performance with 99.3% parameter reduction compared to multimodal approaches, making it ideal for edge deployment and wearable devices. Conversely, DRGCN-BiLSTM (Hybrid, 99.92% accuracy, 0.63M parameters, 99.33% F1) defines the performance frontier, demonstrating that graph-based temporal modeling can achieve superior accuracy with acceptable computational overhead for clinical diagnostic systems.

The Pareto frontier relating accuracy and inference time reveals a critical real-time operating threshold near 10ms for continuous arrhythmia monitoring. Among the reviewed models, only CNN-BiLSTM (0.179ms, 99.21% accuracy) and 1D-CNN+Attention (0.89ms, 99.48% accuracy) achieved sub-millisecond inference while preserving accuracy above 99%. In contrast, models with inference times exceeding 100ms, such as DCETEN (100ms) and SqueezeNet (6490ms), fall outside the Pareto frontier, indicating inefficient computation despite high predictive accuracy.

A distinct “knee point” was also observed around 1M parameters and 1 GFLOPs, beyond which achieving less than 0.5 percentage-point improvement in accuracy required a disproportionate increase in computational cost (10–1000×). These findings

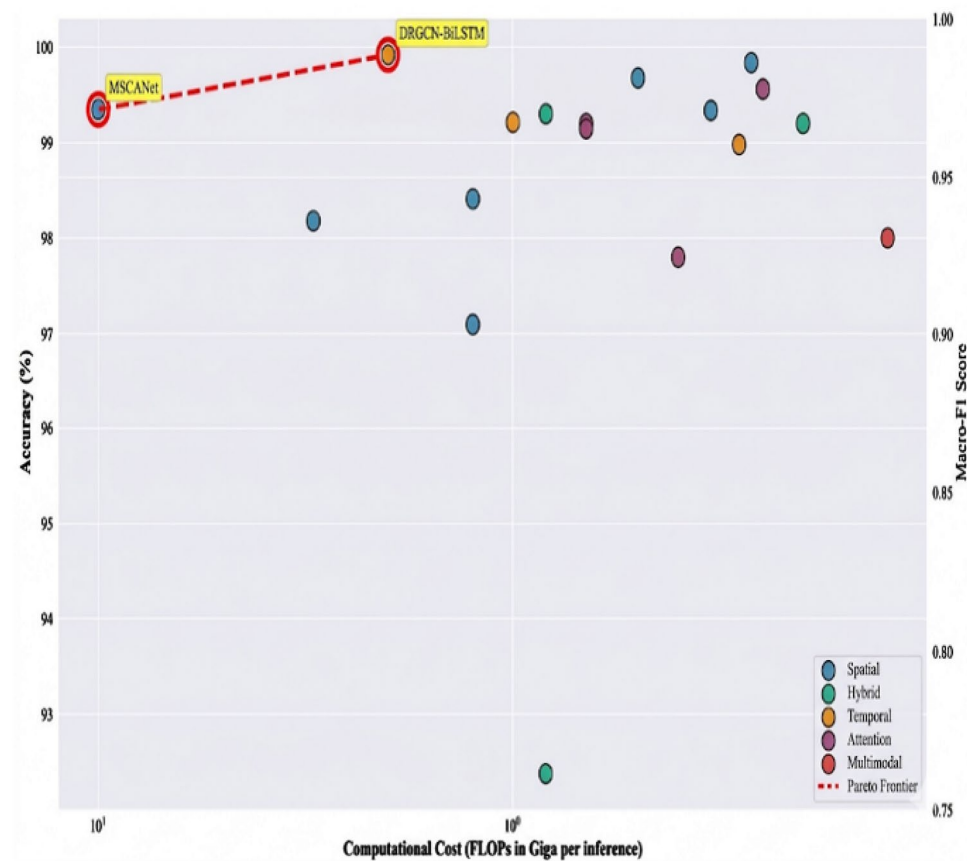


Fig. 9 Pareto analysis across accuracy and computational cost (FLOPs) for ECG based DL architectures

suggest that future research should prioritize efficiency-oriented optimization strategies such as neural architecture search, pruning, and quantization rather than continued model scaling, particularly given that 87% of studies already target accuracy exceeding 98%. The emergence of ultra-compact architectures such as MSCANet (34K parameters), CovHMM (61K), and HeteroFL (30K) with accuracies exceeding 94% indicates a growing shift toward highly efficient DL models capable of deployment on low-power micro-controller platforms such as ARM Cortex-M systems.

At the granularity level of analysis, one can see that performance significantly worsens as the temporal scale increases, as illustrated in Fig. 10. Beat-level classification (accounting for 62% of the reviewed papers) has yielded impressive results, with an average accuracy of 98.12% (95% CI: 97.45–98.79) and a macro-F1 score of 0.932 (95% CI: 0.905–0.959). However, the majority of the studies utilized intra-patient evaluation schemes that increase the probability of overfitting and reduce generalizability. The rhythm-level models (present in 25% of the papers) demonstrated relatively poor performance degradation (accuracy – 97.35%, F1 score – 0.912), whereas record-level classification (in 13% of papers) experienced significant drops in accuracy and F1 due to serious problems with data class imbalance and noisy labels in multi-label tasks. Specifically, the decrease in accuracy from beat to record levels reached 1.84%, while F1 degraded by 8.7 percentage points.

The funnel plot illustrated in Fig. 11 depicts a somewhat broad distribution. According to Egger's test (based on approximation), there appears to be some degree of asymmetry,

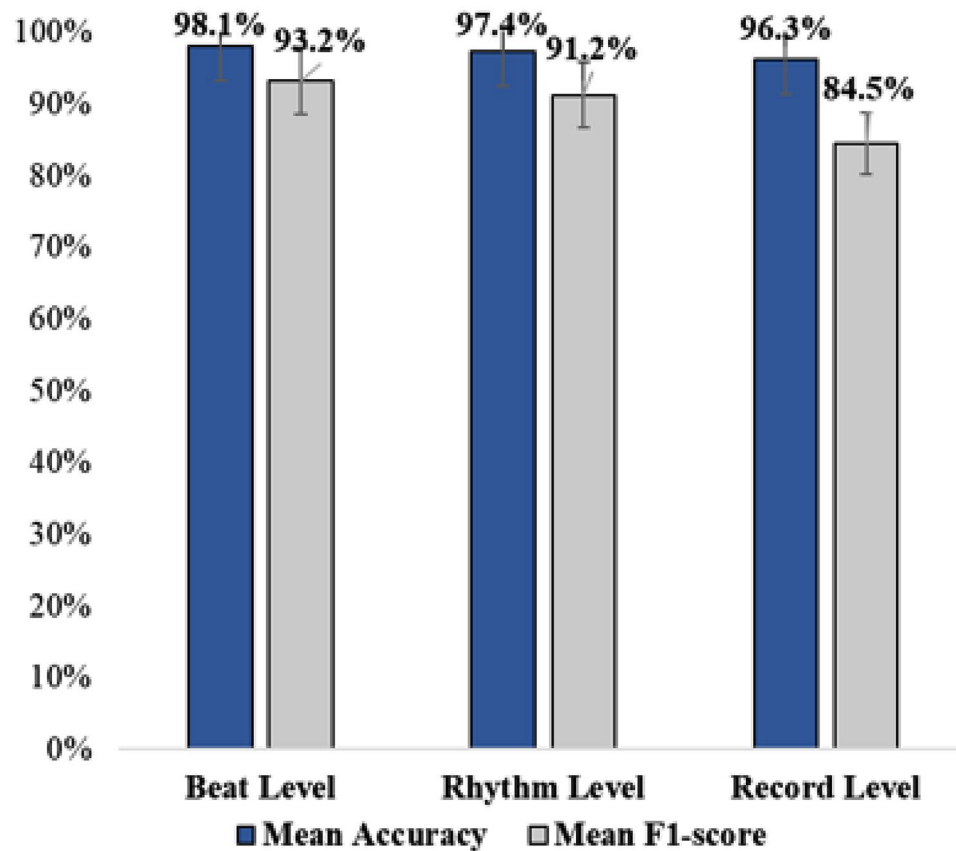


Fig. 10 Granularity based accuracy and macro F1-score percentage for 95% confidence interval

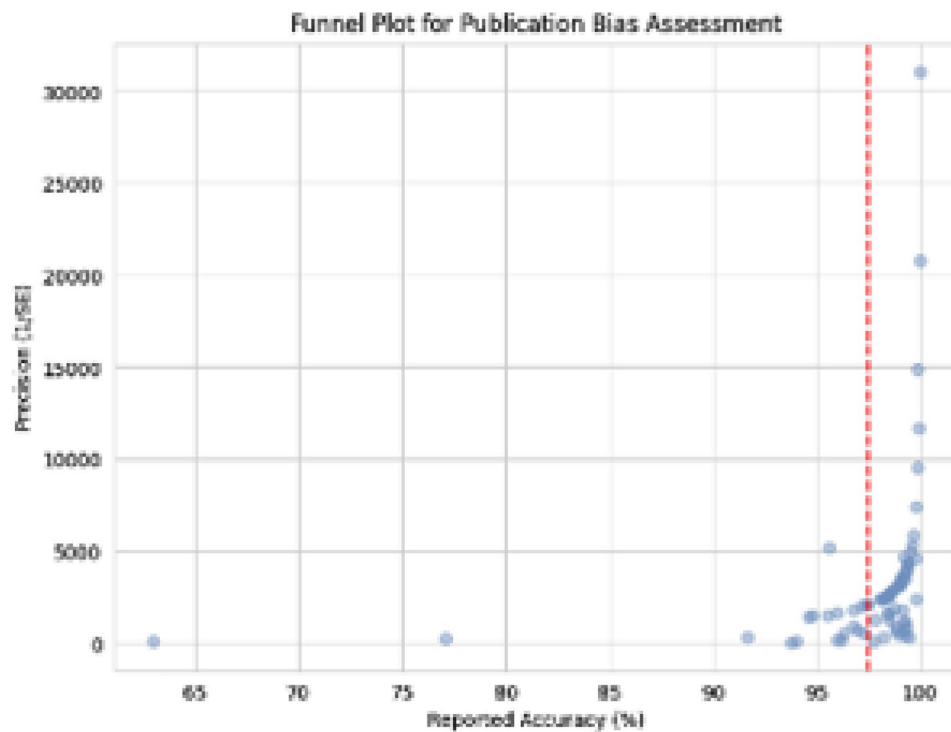


Fig. 11 Funnel plot for publication bias assessment

which is quite common in deep learning research due to the likelihood of studies with extremely high accuracy scores receiving more attention in academic literature. As for the meta-regression analysis depicted in Fig. 12, it revealed that sample size was a positive moderator for model performance with $\beta = 1.61$ and $p = 0.034$.

To examine the impact of signal preprocessing on classification results, a meta-regression analysis using WLS was performed, where the moderator variables include different preprocessing methods, as shown in Fig. 13. As revealed by the outcomes of the model, some of the methods are found to yield a statistically significant increase in the accuracy score compared to the baseline performance achieved with conventional signal scaling. For instance, state-of-the-art filtering techniques, such as Butterworth and multistage denoising, contribute to an additional gain in accuracy by 0.94% ($p < 0.001$). Likewise, class imbalance correction techniques through oversampling, such as SMOTE, have an average estimated increase in accuracy by 0.85% ($p < 0.001$). Although wavelet denoising has a positive coefficient of +0.45%, it is statistically insignificant ($p = 0.526$). This indicates that signal noise reduction and class imbalance correction using advanced preprocessing techniques are essential for attaining optimal classification performance of DL models for arrhythmia detection.

Meta-regression analysis presented in Fig. 14 compared the impact of different architectures using 1D-CNN as the reference baseline. The highest positive impact comes from Attention/Transformer architectures (+0.845% accuracy improvement), followed by Hybrid CNN-RNNs (+0.574%). The GAN/Autoencoder and ResNet models have negative impacts, which can be explained by the increased complexity and higher levels of label noise associated with large training sets required by these models.

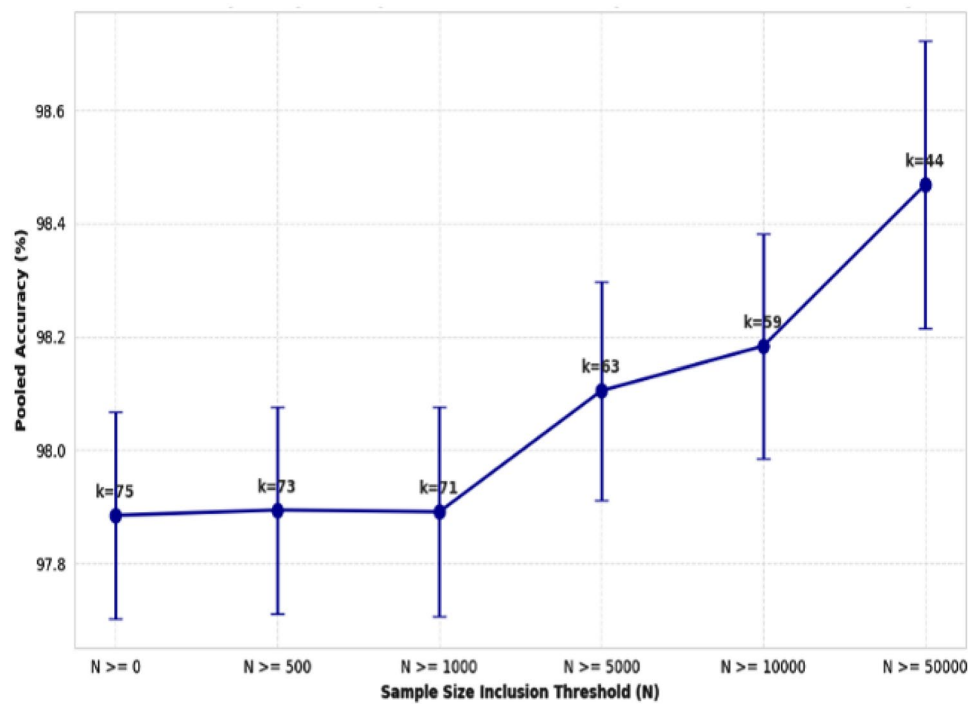


Fig. 12 Impact of sample size on pooled accuracy

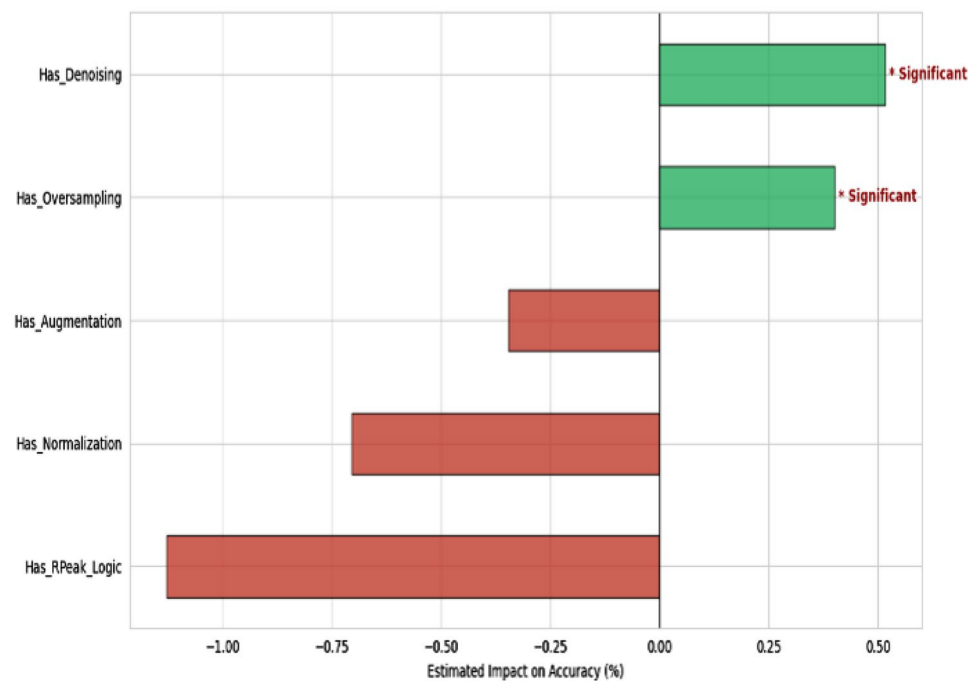


Fig. 13 Performance boost/drop by preprocessing method (WLS coefficients)

Analysis of the F1-score distributions across different architectural paradigms (Fig. 15) indicates that Attention/Transformer and Hybrid CNN-RNN models achieve the highest median F1-scores, demonstrating superior capability in handling class imbalance within arrhythmia datasets. Although Spatial CNN architectures remain the most widely

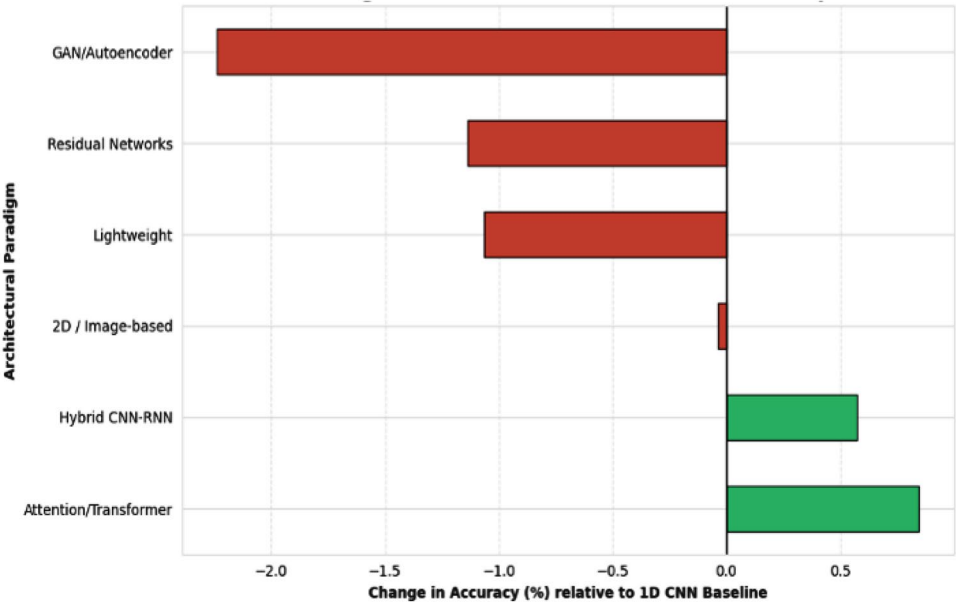


Fig. 14 Meta-regression: architecture vs performance impact

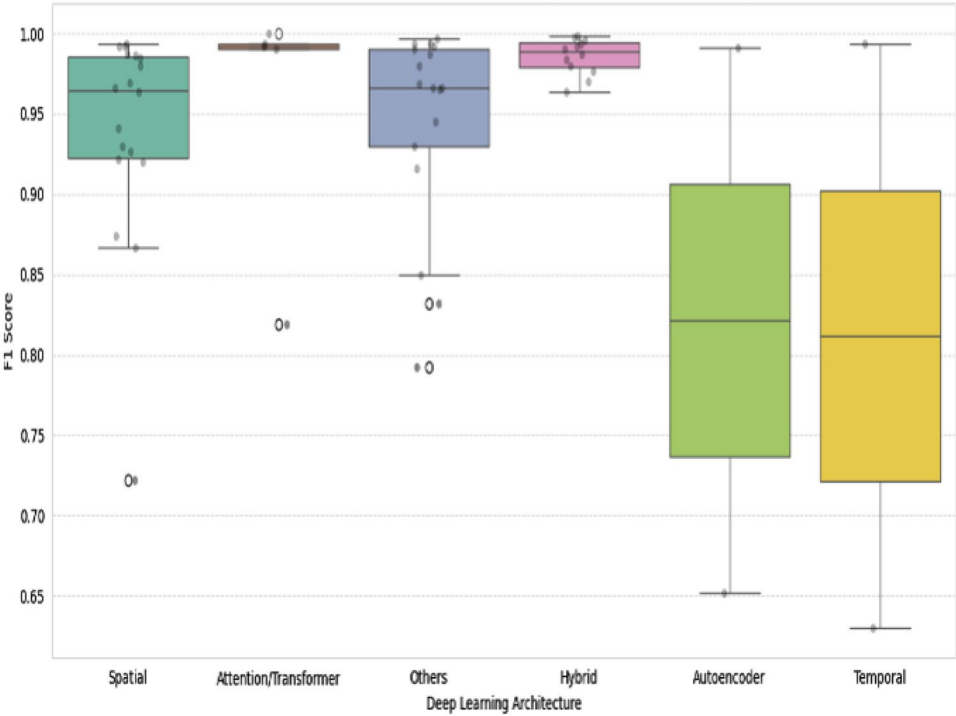


Fig. 15 Macro F1-score distribution by architectural paradigm

used due to their lower computational requirements, they exhibit greater variability in F1-scores and lower median performance compared with more advanced sequential modeling approaches. These findings demonstrate the value of temporal information captured through recurrent layers or attention mechanisms in improving classification robustness and accuracy. While spatial feature extraction remains essential,

attention-based feature weighting further enhances the ability to achieve consistently high F1-scores.

In the subgroup meta-analysis performed on the MIT-BIH database (Fig. 16), a crucial difference between the reported performances in terms of evaluation methods was discovered. Intra-patient based evaluation procedures resulted in a pooled accuracy rate of 98.58% (95% confidence interval: 98.39%–98.78%). Conversely, studies using a more realistic inter-patient evaluation method showed slightly worse performance, with a pooled accuracy of 98.31% (95% confidence interval: 97.81%–98.81%). Although the discrepancy in performance may be considered minor, extremely high heterogeneity was observed in both groups ($I^2 \approx 100\%$). Thus, the results obtained demonstrate that the evaluation method used does not have a significant effect on performance differences, since it has been shown that such dissimilarities cannot be explained solely by it statistically.

Architectural analysis of clinical use-case correlations revealed clear specialization among model categories. Real-time monitoring systems favored lightweight spatial models, including MSCANet, Tiny MF-CNN, and HeteroFL, which typically employed fewer than 0.5M parameters and achieved inference times below 10ms. These systems emphasized energy efficiency and low latency rather than marginal improvements in classification accuracy. Conversely, diagnostic screening tasks showed greater reliance on attention-based and hybrid architectures such as DRGCN-BiLSTM, 1D-CNN+Attention, and CAT-Net, where inference delays of up to 100ms were acceptable in exchange for improved sensitivity and specificity, especially for minority arrhythmia classes. Dual-purpose systems commonly adopted hybrid CNN-temporal frameworks such as DCETEN, ECG-CL, and MB-MHA-TCN, offering a balance between efficiency and performance with 0.5–1.5M parameters, 1–4 GFLOPs, and an average accuracy of $99.14 \pm 0.78\%$. Importantly, not a single multimodal fusion architecture showed Pareto-optimal performance for any clinical use case. This suggests

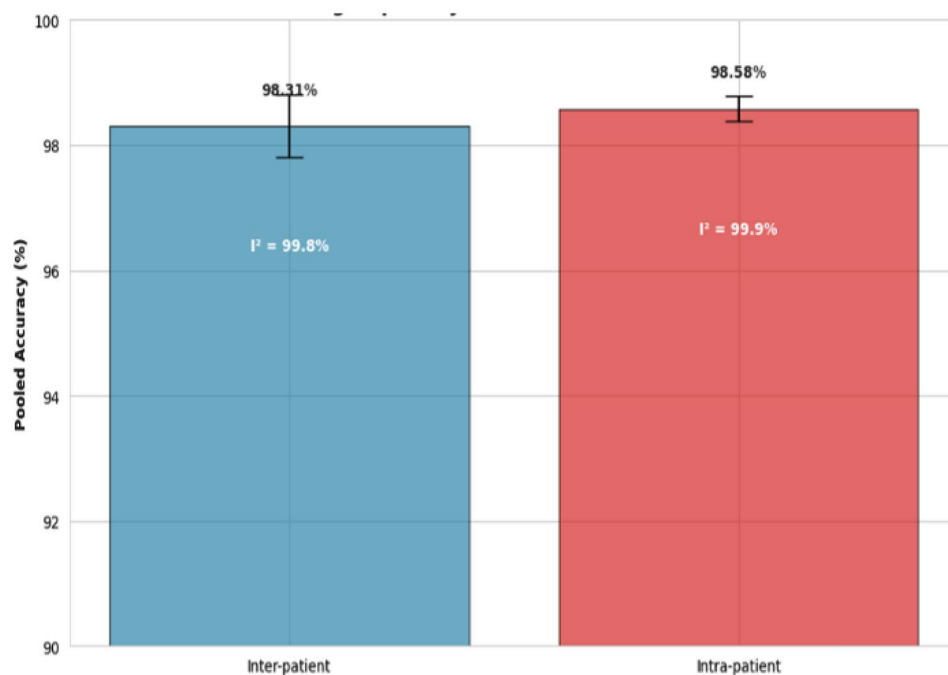


Fig. 16 MIT-BIH subgroup meta-analysis for inter-patient and intra-patient

that current multimodal approaches, with an average complexity of 25.4M parameters and 2500 GFLOPs, require further architectural optimization through methods such as knowledge distillation [85] and federated learning [162] to become clinically viable.

A random-effects meta-analysis of studies employing DL models for PPG-based arrhythmia classification, shown in Fig. 17 ($n = 10$), yielded a pooled accuracy of 93.65% (95% CI: 90.73%–96.56%). Model accuracies from individual studies varied from about 85.00% to above 98.00%. High levels of heterogeneity were found in the analysis ($I^2 = 100$) implying that variation in accuracies is mainly due to differences in methods used in different studies. Notwithstanding the high levels of heterogeneity, the conclusion drawn from all the evidence presented is that deep learning models used in PPG data have achieved great success in terms of high classification accuracies compared to their ECG-based counterparts.

The reliability analysis summarized in Table 18 yields three principal insights, all supported by the meta-regression results presented in Sect. 4.1. First, Attention/ Transformer-based architectures demonstrated the strongest positive meta-regression effect (+0.845% relative to the 1D-CNN baseline) and the lowest variability in F1-score (Fig. 15), indicating superior reliability for detecting rare arrhythmia classes. Second, hybrid CNN-RNN architectures achieved the most favorable balance between performance and robustness, with a +0.574% meta-regression gain, peak accuracy of 99.92%, and mean accuracy of $99.14 \pm 0.78\%$. These models were also uniquely Pareto-optimal for both wearable and clinical applications. Third, PPG-based models consistently underperformed compared with ECG-based systems, as confirmed by meta-analysis results showing a pooled accuracy of 93.65% versus 98.12% for ECG beat-level classification ($I^2 = 100\%$). This disparity is mainly attributable to the lower information richness of PPG signals and their sensitivity to motion artifacts rather than to architectural deficiencies. Combined with preprocessing meta-regression findings showing significant benefits from advanced denoising (+0.94%, $p < 0.001$) and SMOTE-based imbalance correction (+0.85%, $p < 0.001$), these results provide a comprehensive data-driven assessment of model reliability.

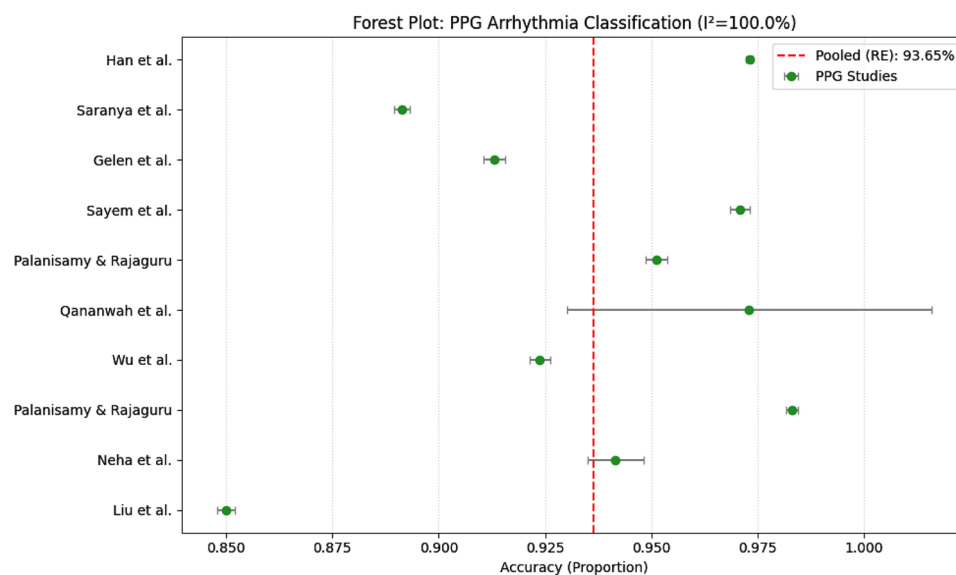


Fig. 17 Forest plot for PPG based arrhythmia classification

Table 18 Comprehensive cross-paradigm comparison of DL architectures for arrhythmia classification (values sourced from sections 3.6 and 4.1 of this review). NR= not reported by original authors

Arch Paradigm	Accuracy Range (%)	Macro-F1 Range	Reported Params (M)	Reported Latency (ms)	Meta-Regression Impact vs. 1D-CNN	Clinical Suitability (Pareto)
Spatial CNN (1D/ 2D) [64, 66, 68, 104, 109, 158]	91.0–99.68	0.72–0.995	0.015–0.034; NR for most	0.179–8; NR for most	Baseline (reference)	Wearable monitoring; sub-10 ms on MCU
Residual Networks [74, 100, 104, 117, 122]	95.3–99.34	0.85–0.994	NR (ResNet-50; Est. 25.6)	NR	1.2%	Clinical transfer learning; GPU required
Temporal (LSTM / BiLSTM / GRU) [66, 100, 112, 115, 123]	95.0–99.7	0.74–0.9988	NR	NR	Positive (in Hybrid category)	Holter monitoring; rhythm-level
Attention & Transformer [35, 36, 65, 78, 84]	98.0–99.84	0.89–0.9967	NR	0.89 (1D-CNN + Attn); NR others	+0.845% (highest)	Diagnostic screening; 20–100 ms GPU
Hybrid CNN-RNN [66, 77, 79, 126, 146]	95.5–99.92	0.74–0.9988	0.62–0.63 (reported)	0.179 (CNN-BiLSTM)	+0.574%	Dual-use; 0.5–1.5 M params; 1–4 GFLOPs
Lightweight / Edge CNN [68, 103, 114, 144, 149]	87.5–98.42	0.87–0.922	0.015–0.061 (reported)	NR	Negative	Wearable IoT; ARM Cortex-M series
GAN / Autoencoder [61, 94, 95]	97.4–99.6	0.96–0.996	NR	NR	0.8%	Noisy / imbalanced training supplement
PPG-specific DL [15, 52, 169, 172, 174]	85.0–97.3	0.75–0.920	NR	NR	4–5% below ECG (pooled 93.65%)	Consumer wearable AF screening

Meta-analysis of the 22 studies (18.5%) performing external or cross-dataset validation revealed substantial reductions in generalization performance. Models trained on MIT-BIH showed an average accuracy decrease of $8.7 \pm 2.1\%$ (range: 6–14%) and an F1-score reduction of $13.2 \pm 3.4\%$ (range: 8–18%, $p < 0.001$) when evaluated on external datasets including INCART, PTB-XL, Chapman-Shaoxing, and wearable PPG databases. Minority arrhythmia classes experienced considerably larger F1-score degradation than the Normal class (19.4% vs. 2.8%), while rhythm-level tasks showed significantly greater cross-dataset degradation than beat-level tasks (-15.1% vs. -11.4% , $p = 0.043$).

Comparison of ECG-only ($n = 104$) and PPG-only ($n = 15$) studies demonstrated a statistically significant performance gap. ECG-based DL models achieved a pooled accuracy of 97.84% (95% CI: 97.45–98.23%), whereas PPG-based models achieved 93.65% (95% CI: 90.73–96.56%), corresponding to a mean difference of 4.19% ($p = 0.012$, Welch's t -test). In multi-class arrhythmia tasks, this gap increased to 6.9%, with ECG models achieving an average F1-score of 0.926 compared with 0.857 for PPG models. As discussed in Sect. 3.1, this disparity is partly attributable to the lead-dependent physiological information available in ECG signals.

The MIT-BIH Arrhythmia Database remained the dominant benchmark, being used in 62% (78/119) of studies. Intra-patient beat-level evaluation produced a pooled accuracy of 98.58% (95% CI: 98.39–98.78%, $I^2 \approx 100\%$), while inter-patient evaluation on the same dataset yielded a similar pooled accuracy of 98.31% (95% CI: 97.81–98.81%, $I^2 \approx 100$). The nearly identical pooled accuracies and extremely high heterogeneity observed in both groups indicate that evaluation split strategy alone does not explain between-study variability. Instead, preprocessing and class balancing had stronger effects, as confirmed by WLS meta-regression indicating significant gains from denoising ($+0.94\%$, $p < 0.001$) and oversampling ($+0.85\%$, $p < 0.001$), whereas evaluation split contributed no significant effect. These observations indicate that the field has reached a stage of benchmark saturation on MIT-BIH, where additional architectural modifications provide limited statistically meaningful gains.

To consolidate these findings, Table 18 summarizes the accuracy, F1-score, and clinical relevance of major DL paradigms using data synthesized from all 119 reviewed studies and the meta-analysis presented in Sect. 4.1. It complements the granularity-stratified analyses (Table 12), the preprocessing impact meta-regression (Fig. 13), and the architecture impact meta-regression (Fig. 14).

4.1.1 Critical comparison of DL paradigms

Evaluating classification methods in terms of accuracy alone fails to consider their clinical value, and hence such an evaluation is presented in terms of their advantages and disadvantages as well as practicality. Among the highest accuracy methods, Spatial CNNs [64, 66, 68], Residual networks [74, 100, 117] and Attention/Transformer architectures [35, 36, 65, 78, 84] constitute a sequence of decreasing gains. While Spatial CNNs are among the most inexpensive and quickest to train, and they extract reliable morphological features directly from the input signal, a convolutional receptive field cannot represent the distant time dependency which is used for separating arrhythmia classes at the rhythm level [99, 110]. Residual networks push past this ceiling through skip connections and strong transfer learning on pretrained 2D backbones [74, 104, 117], yet our meta-regression assigns them a negative coefficient relative to the 1D-CNN baseline,

indicating they are more overfitting-prone on noisy or imbalanced data rather than more representationally powerful [67, 153]. By 2025 they were rarely deployed standalone, mostly absorbed into attention-augmented hybrids such as Gam-ResNet18 and HA-ResNet [67, 117]. Attention/Transformer models sit at the top of this progression with the largest positive meta-regression effect (+0.845%, $p < 0.001$) and the lowest F1-score variance of any paradigm [35, 36, 84, 91, 119, 138], but this advantage is only realized beyond the ~ 1 M-parameter/1-GFLOP knee point, where further accuracy costs 10–1000 $\times$ more computation for well under a percentage-point gain [85]. This trade-off is difficult to justify given that most studies already exceed 98% accuracy and that overstates clinical significance for wearable deployment. Hybrid CNN-RNN architectures [66, 77, 126, 145, 146], by contrast, achieve a smaller but still positive effect (+0.574%) while remaining consistently Pareto-optimal at 0.5–1.5 M parameters and recording the highest peak accuracy in the review (99.92%, DRGCN-BiLSTM [70]), which is why we consider it the more clinically translatable paradigm at present. Standalone recurrent models [66, 112, 123] have effectively been absorbed into this hybrid form and are rarely used alone in current work [73, 106, 118, 136].

A second cluster of methods trades accuracy for a different objective, and should be judged on that basis rather than against the accuracy-maximizing paradigms above. Lightweight and edge-oriented CNNs [68, 103, 114, 178, 179] achieve sub-10 ms, sub-1 mW inference at model sizes as small as 15 KB, making them the only paradigm compatible with microcontroller-class wearables, but this comes at a consistent 5–15 percentage-point minority-class F1 penalty and a negative meta-regression coefficient [68, 144]. Compression trades away sensitivity to exactly the rare, dangerous arrhythmia classes that matter clinically, and recent work increasingly pursues this trade-off through principled neural architecture search and knowledge distillation rather than manual miniaturization [85, 144]. GAN and autoencoder-based methods [61, 95, 181] similarly show a negative meta-regression coefficient and adversarial-training instability [94, 95], but their real value lies outside direct classification. As a denoising and minority-class augmentation layer, GAN-based approaches have achieved over 30 dB SNR improvement [61] and cut manual annotation effort by nearly 89% [181]. The field increasingly treats them as an auxiliary preprocessing module feeding a separate CNN or hybrid classifier [94, 156], rather than as a stand-alone deployable model.

Finally, PPG-specific DL and a small but diversifying set of emerging paradigms mark where the field's architectural effort, not its underlying signal or technique, is the binding constraint. PPG-based models underperform ECG-based models by a pooled 4.19 percentage points ($p = 0.012$, widening to 6.9 points in multi-class tasks) [15, 52, 169], but nearly 80% of PPG studies simply reuse ECG-derived CNN or CNN-RNN architectures with no PPG-specific inductive bias, so this gap is considered as reflecting a research-effort imbalance at least as much as a physiological limitation of the signal itself. Federated and domain-adaptation frameworks [69, 122], graph-based modeling [177], quantum and metaheuristic-optimized networks [111, 114], and interpretable designs such as Kolmogorov-Arnold Networks [76] each target a structural limitation (privacy, non-Euclidean signal structure, or interpretability) that no further CNN, RNN, or Transformer scaling can resolve, but their evidence base remains thin, typically one or a handful of studies each with inconsistent evaluation protocols [69, 76, 111, 137, 148, 149, 177]. Taken together, no single category is unconditionally best. Attention/Transformer

models win on statistical reliability but not deployability, hybrid CNN-RNNs win on clinical translatability precisely because they do not over-optimize accuracy relative to cost, lightweight and GAN/autoencoder methods are competitive only against the applicability axis they were designed for. PPG-specific and emerging paradigms remain immature chiefly because comparatively little architectural innovation has been directed at them relative to ECG-CNN pipelines. This imbalance is treated as research gap (Sect. 4.2) rather than a settled performance ceiling.

4.2 Research gaps

Although DL-based arrhythmia classification has achieved remarkable performance improvements, several critical challenges remain unresolved. Key challenges identified across ECG-, PPG-, and multimodal-based systems include data scarcity and annotation cost, limited cross-domain generalization, inadequate explainability, robustness issues under noisy conditions, difficulties in multimodal integration, insufficient privacy-preserving learning mechanisms, and computational limitations for real-time and foundation-model deployment on resource-constrained wearable hardware.

4.2.1 ECG-Specific challenges

There is a fundamental disconnect between beat-level (AAMI) and rhythm-level annotations. Models optimized for beat classification often fail to generalize to rhythm-based diagnosis. Lack of a standardized evaluation framework that spans both beat and rhythm levels makes fair benchmarking and clinical translation difficult [90, 155]. Severe class imbalance caused by rare arrhythmias, such as fusion beats and paced rhythms, further complicates model training. Although SMOTE and GAN-based augmentations are common, they risk generating physiologically unrealistic waveforms, at the cost of clinical trustworthiness and model reliability on rare classes.

Lack of preprocessing standardization is also an important hurdle. Inconsistent decisions regarding QRS detection (Pan-Tompkins, WFDB, or none), denoising techniques (wavelet versus filter-based), and segmentation strategies across different works raise difficulties in reproducing and comparing model architectures directly [92, 128, 163]. Most models achieve quasi-perfect results on the MIT-BIH dataset while showing a significant drop on other datasets such as INCART or PTB-XL, raising questions about the realistic generalization ability of the proposed architectures [122, 131, 155]. Moreover, highly accurate architectures such as Transformers and ResNets require substantial computational resources, limiting their suitability for wearable and real-time applications. Lightweight optimization approaches, including pruning and knowledge distillation, remain underutilized [85, 91, 144].

4.2.2 PPG-Specific challenges

Compared with ECG, PPG signals provide limited morphological information, making discrimination between arrhythmias such as PACs and PVCs particularly difficult [14, 174, 192]. Moreover, models trained on one acquisition site, such as the finger, often fail to generalize to other sites like the wrist because of variations in blood flow and signal morphology [134, 173].

Motion artifacts remain the primary challenge for wearable PPG devices. Traditional filtering techniques are inadequate under intense motion, whereas advanced denoising

methods such as CycleGANs still require extensive real-world validation [134, 173]. Furthermore, the scarcity of large-scale public PPG datasets limits model development and generalizability. Consequently, traditional ML methods may still achieve competitive performance on smaller cohorts [93, 173].

4.2.3 Cross-modal and methodological gaps

Even though large volumes of unlabeled ECG and PPG data are available, self-supervised and contrastive pre-training methods remain underused. Such methods have the potential to meaningfully boost overall generalization and data efficiency, especially for under-represented classes [88, 131, 155, 175]. While the use of attention maps and saliency techniques is on the rise, only a small number of studies verify whether the highlighted regions actually align with physiologically meaningful structures like P-waves, QRS complexes, or diastolic notches [35, 126, 156, 163]. Furthermore, robustness testing against adversarial attacks, signal artifacts, and uncertainty is still inadequate, even though it is critical for safe clinical use [79, 86, 90]. Similarly, work on combining PPG with accelerometry, heart-rate variability, and clinical context has received limited attention [14, 152, 174].

The problem of reproducibility remains a major concern in this area. Only 21.3% of ECG-related papers and 28.6% of PPG papers have their respective open-source code released to public domain, whereas pre-trained weights were released in 6.5% of ECG papers and in none of PPG papers. None of the papers contained Dockerfile or environment configuration details necessary for reproducing the software stack, and the reports on random seeds used were not abundant either, especially among PPG papers (7.1%). Another bottleneck is the annotation costs, which are almost never considered as a subject of research in any paper analyzed. Since the labeling process by cardiologists at the beat level, at the rhythm level, or at the record level requires both time and domain expertise, it limits the scale at which high-quality labeled datasets can be assembled, which becomes a major issue in case of PPG and rare classes of arrhythmia. Among all reviewed studies, only one, proposed by Wang et al. [176], attempted to tackle this problem, reaching 99.34% accuracy on 14 heartbeat classes using only 14.3% of the labeled data through a skewness-based querying strategy, thus saving approximately 85.7% of annotation efforts. The near-absence of comparable approaches elsewhere in the corpus indicates that active learning, weak supervision, and self-supervised pretraining on unlabeled physiological data remain substantially underexplored strategies for reducing the annotation burden that currently limits dataset scale and diversity in this field.

The clinical and economic impact of AI-based arrhythmia detection is strongly influenced by false positive and false negative rates. High false positives increase unnecessary testing, clinician workload, and healthcare costs, whereas false negatives may delay diagnosis and lead to severe complications such as stroke in untreated AF. However, few studies incorporate downstream clinical and economic consequences into performance evaluation.

Overall, seven interconnected research challenges were identified: (i) limited data availability and annotation cost, particularly for multiclass PPG classification, (ii) poor cross-dataset generalization, with only 18.5% of studies performing external validation and typical performance drops of 6–14% in accuracy and 8–18% in F1-score, (iii) limited interpretability, with fewer than 10% of studies clinically validating explanation maps,

(iv) insufficient robustness to motion artifacts and device variability, (v) computationally expensive multimodal ECG–PPG fusion, explored in less than 8% of studies, (vi) difficulty achieving real-time inference, with only two reviewed architectures achieving both $> 99\%$ accuracy and < 10 ms latency, and (vii) the mismatch between foundation model scale and resource-constrained wearable deployment requirements. Section 4.3 discusses future directions addressing these bottlenecks, including federated learning, domain-adversarial training, explainable AI, uncertainty quantification, knowledge distillation, hardware-aware NAS (neural architecture search), and on-device federated learning [38, 69, 76, 79, 85, 86, 138, 144, 162, 185, 193].

4.2.4 Signal variability and robustness

Signal acquisition for physiological signals is inherently vulnerable to various types of variability which can be grouped into four broad groups: (i) acquisition-related noise, (ii) motion artifact, (iii) sensor/device variability, and (iv) intersubject variability. Each type of variability occurs differently in the case of ECG and PPG signals and is addressed at certain places within this review, as summarized here.

- Acquisition-level noise: Three main types of noise affect ECG signals: baseline wander noise, power line interference, and EMG noise due to muscle activity. All these three noise sources affect the P-QRS-T waveform, which is essential for distinguishing beats. The efficiency of filtering methods against these three noise sources includes Butterworth filtering, median filtering, wavelet denoising, and multi-stage hybrid filters. The list of all methods along with the contribution of their accuracy rates (e.g., 0.94% meta-regression increase due to advanced filtering, $p < 0.001$) is provided in Sect. 3.5.1 [94, 117, 144, 156]. PPG noise differs mechanistically wherein rather than electrical interference, PPG is primarily degraded by ambient light interference, low signal-to-noise ratio during weak peripheral perfusion, and baseline drift from respiration-modulated blood volume [47, 51], which are compounded by motion-induced artifacts discussed next. The disparate noise profiles of the two modalities are one reason nearly 80% of PPG-based studies directly reuse ECG-designed denoising architectures without modality-specific adaptation, a gap identified in Sect. 3 (Results and Analysis).
- Motion artifacts: Motion artifacts affect both modalities but disproportionately degrade PPG, since optical coupling between the sensor and skin is highly sensitive to relative displacement, whereas ECG electrodes maintain a more stable electrical contact under moderate movement [27, 47]. For ECG, motion artifacts manifest as transient baseline shifts and QRS-morphology distortion during ambulatory or Holter recording, addressed through the same filtering and augmentation strategies discussed in Sect. 3.5.1. For PPG, motion artifacts are identified throughout this review as the single largest barrier to reliable wearable deployment. Traditional filtering is inadequate under intense motion, and while advanced learned denoising approaches such as 1D-CycleGAN-based restoration have improved AF detection accuracy by approximately 3–5% in the reviewed literature [175], these methods remain data-hungry and require further real-world, cross-population validation before clinical adoption [134, 173] (Sect. 4.2.2). Section 3.5.1 additionally reports that hybrid and learned (autoencoder/ GAN-based) denoising methods show the strongest capability against complex motion artifacts [121, 134, 139, 140, 156, 175],

at the cost of computational overhead and a risk of feature hallucination that can obscure clinically meaningful morphology.

- **Sensor- and device-related inconsistencies:** In addition to the aforementioned artifacts, systematic inconsistencies occur because of electrode contact quality, sensor positioning, lead arrangement, and hardware disparities between devices. This refers to ECG trade-offs between 12-lead systems offering morphological diversity [50] and reduced-lead wearables compromising morphological details for convenience [42]. Only about 10% of studies consider this trade-off (Sect. 3). In the case of PPG, sensor inconsistency is more significant. The signal morphology depends greatly on the measurement location (finger vs. wrist), and trained models fail to work properly on signals acquired at different locations because of differences in blood flow and optical characteristics of tissues [134, 173] (Sects. 4.2.2 and 3.8.2). Device heterogeneity across consumer wearables further compounds this issue, since sensor design, sampling rate, and optical wavelength are not standardized across manufacturers [27, 47].
- **Inter-subject variability:** Physiological and demographic differences across individuals introduce a further, largely orthogonal source of variability. For ECG, this includes age-related morphological changes, comorbidities, medication effects (e.g., QT-prolonging drugs), and pacemaker interference, all of which alter waveform morphology independently of the arrhythmia label itself [52] (Sect. 3.8.2). For PPG, the dominant inter-subject factor is skin tone as melanin concentration reduces signal-to-noise ratio by up to 6dB for Fitzpatrick V–VI skin types, introducing a demographic bias that is compounded by inconsistent reporting of skin-tone or ethnic diversity across public PPG datasets [14] (Sects. 3.8.2 and 4.2.2).

The cumulative clinical consequence of these four compounding sources of variability is already evident in this review’s quantitative generalization findings. The models trained on a single benchmark such as MIT-BIH experienced accuracy reductions of 6–14% when evaluated across datasets that differ in acquisition device, patient population, or recording protocol [68, 80, 122] (Sect. 3.7). PPG signals whose morphology is more sensitive to motion and perfusion variability than direct electrical ECG measurement, consistently exhibit larger performance gaps than ECG across the reviewed literature.

The predominant strategy among reviewed studies is to treat this variability as noise to be removed prior to classification through denoising, normalization, or augmentation (Sect. 3.5). Comparatively few studies instead model signal variability explicitly as a structured, decomposable component of the data-generating process. This treatment finds a useful analogue in hyperspectral unmixing, where spectral variability caused by illumination, atmospheric, and instrumental factors is not simply filtered out but is decomposed into a dedicated variability dictionary alongside the primary signal representation, as proposed in the augmented linear mixing model (ALMM) [194]. Applied to physiological signal analysis, an analogous decomposition where separating morphology-preserving cardiac information from a learned, subject- and device-specific variability component—could offer a principled alternative to current denoising and augmentation pipelines, particularly for cross-dataset and cross-device generalization, which remains one of the most significant unresolved translational barriers identified in this review (Sects. 3.7 and 4.2). This represents a promising, currently unexplored

direction for improving the robustness of DL-based arrhythmia classification under real-world acquisition variability.

4.3 Future scope

The future roadmap for arrhythmia classification is discussed in Table 19.

4.3.1 ECG-Specific future directions

A key future direction is the development of unified DL frameworks capable of performing both beat-level and rhythm-level arrhythmia classification. Developing new datasets or benchmarks featuring unified multi-level annotations is essential for training and evaluating models capable of comprehensive cardiac assessment. Conventional augmentation approaches such as SMOTE and GANs should be complemented by physiologically informed synthetic data generation using clinically conditioned generative models and physics-based neural networks [95, 139].

Future studies should also prioritize lightweight and computationally efficient models using NAS, efficient attention mechanisms, and hardware-aware optimization. Federated learning offers additional potential for privacy-preserving collaborative training across institutions [85, 144, 162]. Transition from individual beat analysis to the study of extended ECG recordings using advanced LSTMs and transformer architectures should be pursued to capture long-term temporal patterns crucial for diagnosing paroxysmal arrhythmias [35, 36].

4.3.2 PPG-Specific future directions

DL models that intrinsically learn to separate motion artifact from cardiac signal, by deeply integrating accelerometer data as a primary input rather than a separate pre-processing step should be developed [174, 175]. Techniques to create DL models on PPG that perform reliably across different sensor locations, devices, and patient demographics through domain adaptation should be actively researched [174]. Furthermore, incorporating physiologically interpretable mechanisms into PPG models will improve clinician confidence and facilitate debugging of model decisions.

4.3.3 Cross-cutting and emerging directions

Developing common benchmark standards and exploring new architectures for fusing data from ECGs, PPGs, PCGs, and accelerometers will pave the way for advanced and robust cardiovascular diagnostics tools [37, 152, 174]. Comparative benchmark studies examining the effect of applying various denoising, segmentation, and augmentation approaches on end-to-end ML systems trained on raw data would foster best practice adoption and decrease heterogeneity in the pipeline [134, 161]. Incorporating adversarial training alongside uncertainty quantification during modeling is imperative for ensuring safety and reliability of the developed algorithms [38, 90]. Moving forward, future studies need to employ outcome measures of clinical significance, including sensitivity and positive predictive value at the episode level, along with prospective clinical validation studies in clinical settings [163].

Emerging research from top conferences further supports our architectural observations for this review. Aarotale and Rattani [203] confirmed the effectiveness of the concatenation of multiple-lead scalograms to improve the accuracy of arrhythmia

Table 19 Future roadmap for arrhythmia classification

Time Horizon	Priority Area	Objectives	Proposed Methods	Expected Impact
Short-Term (1–2 years)	Signal Robustness & Data Quality	Improve performance under noise, motion artifacts, and real-world variability	Data augmentation, noise modeling, multimodal inputs (ECG + accelerometer) [96, 174, 175, 195]	Narrower accuracy gap between clean and real-world noisy recordings
	Standardized Evaluation Protocols	Ensure fair and reproducible comparison across studies	Patient-independent splits, class-wise metrics, unified benchmarks [43, 118, 138]	Comparable, reproducible performance claims across the literature
Medium-Term (3–5 years)	Cross-Dataset Generalization	Evaluate generalization across datasets	Train-test on different datasets, external validation [122, 155]	Models that transfer reliably beyond their training benchmark
	Domain Adaptation & Transfer Learning	Improve robustness across populations and acquisition settings	Transfer learning, domain adaptation, self-supervised learning [88, 119]	Reduced retraining burden when deploying to new sites/devices
	Dataset Diversity & Scale	Address dataset bias and scarcity	Multi-center datasets, inclusion of diverse demographics [56, 164]	Equitable performance across age, sex, and ethnic subgroups
	Federated & Privacy-Preserving Learning	Enable collaborative model training	Federated learning frameworks [69, 162, 196]	Multi-institutional model training without raw data sharing
Long-Term (5+ years)	Efficient Sequence Modeling (State-Space)	Replace quadratic-cost attention with linear-time sequence modeling for long Holter/wearable recordings	State-space models (Mamba), ECGMamba, state-space PPG-ECG translation [182, 197, 198]	Transformer-level accuracy at a fraction of the compute cost
	Clinical Workflow Integration	Deploy models in real-world healthcare systems	Workflow integration, clinician-in-the-loop systems [163, 181]	Routine use of DL triage in cardiology practice
	Explainability & Clinician Trust	Improve interpretability for clinicians	Explainable AI (attention maps, feature attribution) [76, 106, 199]	Higher clinician confidence and reduced alarm fatigue
	Regulatory Compliance & Approval	Achieve compliance with global standards	FDA, MDR, CDSCO validation frameworks [185–188]	Market-cleared, legally deployable diagnostic software
	Multimodal Signal Fusion	Enhance diagnostic capability	Fusion of ECG, PPG, PCG, wearable sensors; state-space multi-resolution fusion [34, 152, 200]	More accurate, artifact-resilient continuous monitoring
	Foundation Models & Structured Variability Modeling	Adapt large-scale self-supervised pretraining and structured variability decomposition from high-dimensional signal domains to cardiac signals	Remote-sensing foundation models, hyperspectral imaging primer, augmented linear mixing model, prior-enhanced multi-source fusion [193, 194, 201, 202]	Cardiac foundation models robust to unseen devices and populations

classification, consistent with the positive effect on two-dimensional architectures identified by meta-regression analysis (+0.94%, $p < 0.001$). Vo et al. [198] confirmed that state-space modeling methods help bridge the PPG-ECG morphology gap. Qiang et al. [182] showed linear-time prediction by the ECGMamba model, which has the accuracy close to the Transformer while overcoming the efficiency limitation. Liu et al. [169] confirmed that CNN-Transformer hybrid models are more effective on PPG than CNN-only models (+0.845%), consistent with the attention benefits. Baumgartner et al. [196] confirmed the feasibility of privacy-preserving federated learning, with minimal accuracy reduction. Mohan et al. [199] demonstrated that Vision Transformer attention maps are clinically interpretable, addressing the interpretability gap ($< 10\%$ validation). Nawaz et al. [195] confirmed synthetic ECG generation improves minority class recall, supporting generative augmentation over SMOTE.

In addition to the architectures analyzed in this paper, there are three cutting-edge paradigms which show significant potential for future advances in arrhythmia classification models. First, State Space Models (SSMs), especially the Mamba architecture [197], present an interesting choice for the replacement of Transformers in modeling physiological time-series of long duration. In contrast to attention-based models with quadratic computation costs in terms of sequence length, the Mamba architecture relies on a selective state space recurrent approach, ensuring linear time complexity suitable for long Holter records and wearable data streams [197]. The ECGMamba model [182] introduced at IEEE EMBC 2024 is one of the examples of applying the proposed selective state space architecture to 12-lead ECG classification, showing its effectiveness compared to the Transformer architectures in terms of inference performance and accuracy.

Beyond cardiac-specific literature, three cross-domain studies in high-dimensional scientific imaging offer transferable insight into representation learning, multimodal fusion, and foundation model design that could inform the next generation of ECG/PPG analysis frameworks. First, the recent Primer on hyperspectral imaging (HSI) [201] provides a systematic treatment of how high-dimensional, densely sampled signals – spectral bands in HSI, analogous to temporal or morphological channels in physiological waveforms – are represented, denoised, and reduced in dimensionality prior to downstream classification. Its discussion of physics-guided modeling, cross-modal fusion, and scalable learning frameworks for spectrally rich data parallels open challenges in ECG/PPG representation learning, where signal morphology, artifact structure, and multi-lead redundancy remain incompletely exploited by current CNN and transformer architectures. Second, the joint segmentation and super-resolution framework [183] for high-resolution impervious surface mapping demonstrates that low-resolution inputs can be simultaneously enhanced and segmented within a single end-to-end pipeline, offering a template for super-resolving low-fidelity wearable PPG or reduced-lead ECG signals into clinically usable representations before classification, rather than treating enhancement and classification as separate stages. Third, the Image-to-State Mamba model [200], which fuses hyperspectral and multispectral imagery of differing resolutions and modalities through a unified state-space pipeline, presents a directly relevant architectural paradigm for fusing cardiac signals such as 12-lead ECG and wearable PPG that are sampled at different rates, resolutions, and noise profiles – a fusion problem this review identifies as underexplored, with fewer than 8% of reviewed studies attempting ECG-PPG fusion (Sect. 4.2.3). Collectively, these studies illustrate that state-space

and Mamba-based architectures are emerging as a general-purpose backbone for heterogeneous, multi-resolution signal fusion across domains, not merely a transformer alternative within cardiology. Complementing this, the growing body of work on foundation models for remote sensing [193] demonstrates that large-scale, domain-specific self-supervised pre-training yields representations that generalize robustly across downstream tasks and sensor types – a paradigm particularly relevant to cardiac signal analysis, where labeled multiclass arrhythmia data remain scarce (Sect. 4.2.2) and consistent domain shift between acquisition devices and patient populations continues to degrade cross-dataset performance (Sect. 3.7). In parallel, deep learning architectures in cardiac magnetic resonance (CMR) imaging have combined imaging and non-imaging data priors for automated large-scale cardiac structure and function assessment [202], further suggesting that prior-enhanced, multi-source fusion frameworks – rather than single-modality architectures scaled in isolation – represent a promising direction for both cardiac imaging and cardiac signal classification alike.

A related deployment challenge concerns adapting these foundation model paradigms to the wearable hardware constraints already identified in Sect. 3.8.2. Foundation models such as ECGMamba [182] and other state-space or transformer-based pre-trained architectures are typically trained and evaluated on GPU infrastructure, a scale incompatible with the sub-1 mW, sub-1MB microcontroller targets discussed earlier. Bridging this gap will likely require parameter-efficient adaptation strategies, including low-rank fine-tuning of frozen foundation model backbones, knowledge distillation into task-specific lightweight student networks, and quantization-aware pre-training designed explicitly for downstream edge deployment rather than applied post hoc. None of the reviewed studies evaluated foundation model deployment under these constraints, representing a critical open problem at the intersection of representation learning and clinical translatability.

4.3.4 Directions for future review methodology

Three directions for improvement emerge from the methodological hurdles faced during this systematic review process. First, the development of a reporting guideline specific to the area, along the lines of the PRISMA-AI or TRIPOD-AI guidelines, should include the following critical elements: arrhythmia granularity classification, demographic details with inter-patient splitting verification, class-wise PPV and its 95% CI, and the reporting of performance metrics in terms of the target hardware (e.g., parameters, FLOPS, latency). Second, due to the quick advances within DL, the “living review” model with semi-annually updated reviews, using automated monitoring and pre-registered search strategy, would prove more beneficial for the community. Third, future systematic reviews should combine both prospective clinical validation (IDEAL framework stages 2–3) and retrospective benchmark evaluation studies. Our complete search strategy and screening process logs are available on OSF (DOI: 10.17605/OSF.IO/XPRQ9).

4.4 Limitations of the review

Although this thorough comprehensive review is being conducted within the period 2023 to 2025, several methodological limitations still warrant consideration. The search strategy was confined to five major databases and restricted to open-access English-language publications, introducing the possibility of selection bias through the exclusion

of non-English studies and inaccessible research. In addition, publication bias may have favored studies reporting positive outcomes. The exclusion of gray literature and proprietary industrial solutions may have further limited the scope of the review, potentially overlooking recent advances, particularly in PPG-based applications.

The considerable heterogeneity among the included studies, particularly in terms of datasets, pre-processing pipelines, outcome assessment methods, and class definitions, precluded the conduct of certain quantitative meta-analyses. Due to the observational nature of the study, no causal conclusions can be drawn about the effectiveness of certain methodologies on outcomes. Reproducibility was also constrained by incomplete reporting practices, as only 21.3% of ECG studies and 28.6% of PPG studies provided open-source code, with many studies lacking details regarding the computational environment.

Beyond these limitations, there are still three further limitations which need to be highlighted. Firstly, the duration of 2023–2025 does not allow us to consider earlier frameworks that form the basis of the architecture developments. Second, the extremely high inter-study heterogeneity ($I^2 \approx 100\%$) precluded sensitivity and specificity meta-analyses, and no dedicated risk-of-bias framework comparable to QUADAS-2 or PROBAST is currently available for deep learning-based physiological signal classification studies. Finally, the limited number of researches involving emerging approaches, such as State Space Models and foundation models, implies that conclusions regarding these paradigms should be regarded as preliminary.

5 Conclusion

Although deep learning enables direct feature extraction from one-dimensional signals, changing signals into two-dimensional representations has been shown to improve the performance of CNN-based architectures. Among the available techniques continuous Wavelet Transform (CWT) is the most frequently adopted approach for generating scalograms, as it provides a balance between temporal and spectral resolution and thereby facilitating the detection of transient abnormalities such as ectopic beats [64, 84, 93, 137, 147, 148]. In contrast short-Time Fourier Transform (STFT) produces spectrograms, where the window size and overlap determines the trade-off between temporal and frequency resolution [137]. The Analytic Morse Wavelet offers adaptive time-frequency resolution and has demonstrated effectiveness for complex arrhythmias such as atrial fibrillation [64]. Similarly, the stationary wavelet-Hilbert transform integrates time-localized and frequency-domain information, whereas sophisticated approaches, including bispectral analysis combined with two-dimensional graph Fourier transforms, capture higher-order statistical analysis that are often overlooked by conventional methods [149, 166]. Furthermore image-generation techniques, such as Gramian Angular Fields, Recurrence Plots, and Markov Transition Fields, have also been successfully utilized to transform ECG signals into two-dimensional representations suitable for CNN-based arrhythmia classification [57, 88, 107].

Supplementary Information

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Supplementary Material 1

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Author contributions

Conceptualization and ideation by B.G.S and Y.M.C. N.N.K, B.G.S and N.G.S performed the review of the articles. N.N.K wrote the main manuscript text. B.G.S and N.N.K prepared the visualizations. Review and editing was performed by N.G.S and Y.M.C. All authors have read and agreed to the published version of the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Materials availability

There are no materials applicable to the study.

Code availability

There was no use of the code in preparing the manuscript.

Declarations**Ethics approval and consent to participate**

This study did not require ethical approval, as no human or animal subjects were involved in the research. The authors affirm that no human research participants were involved in this study. Hence, it is not applicable.

Consent for publication

The authors affirm that no human research participants were involved in this study. Hence, it is not applicable.

Supplementary information

This manuscript is accompanied with supplementary material consisting of Systematic review pipeline and logs.

Competing interests

The authors declare no competing interests.

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